The Dirichlet Pólya Problem for Spherical Shells in General Dimension: Exact Dimension Lifting and the Shifted-Tail Bottleneck

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Abstract

For a spherical shell in dimension $d \geq 3$, the separated radial problem has Bessel orders $\ell + (d-2)/2$ and spherical-harmonic multiplicities $\kappa_{d,\ell}$. We give an exact branching decomposition of the strict shell-action proxy into positive integer- and half-integer shifted planar tails. We also prove a signed primitive comparison and retain its full positive remainder. This yields the exact identity

$$W_d - P_d^< = \mathcal{B}_d(A) + \sum_{m\geq 0} \binom{m+d-3}{d-3} D_A\left(m + \frac{d-2}{2}\right).$$

Consequently, one dimension-free positive shifted-tail inequality implies the sharp Dirichlet Pólya inequality for every spherical shell and every $d \geq 3$. The dimension lift is proved here; the inner shifted-tail inequality remains the central open lemma. We prove the outer and terminal tail regions, close the first interface-cell milestone by exact rational wall estimates, and prove all shifted tails above an explicit fixed-ratio high-energy threshold. A wall-safe first-shelf reduction compresses every remaining many-cell tail to one scalar coupled to its terminal one-interface reserve. In the thin sector it also improves the uniform threshold to $K \geq 125/[8(1-\rho)^2]$. A scale-free shelf-curvature bound and a fractional-top inverse-action estimate give an explicit exact-angle lower bound for the terminal reserve, including the previously unused zero-level triangle. We also prove an automatic ultra-thin high-floor sector and exclude every noncompact escape in both the first-floor and high-floor first-drop branches. A global endpoint-chord and terminal-payment argument first excludes every negative high-floor first drop with $\rho > 99/100$, without a restriction on the frequency or on terminal walls. A normalized positive-arccos refinement first extends this exclusion to $\rho \geq 603/625$. A terminal stratification on the zero-level face then pushes it to $\rho \geq 477/500$. The active-width relation removes the resulting zero-count obstruction, and a sharp terminal-cap estimate extends the exclusion to $\rho \geq 93/100$. A refined absent-level chord and the automatic $q \geq 5/2$ terminal start extend it further to $\rho \geq 927/1000$. Combined with the fixed-ratio scalar estimate, this reduces every residual high-floor candidate to

$$\frac{1}{4626882} < \rho < \frac{927}{1000}, \qquad \frac{21}{4} < K < 2313441.$$

This is a sharper compact reduction, not a proof of the residual fixed-ratio walls. A directed-rounding certificate proves all 19602 finite first-floor chambers with $r \geq 3/2$, $s \in \{0, 1\}$, $x < 100$. A second certificate handles all 398 finite small-start chambers, while an analytic comparison closes their infinite rays. Together with the continuous large-ray certificate, this proves the entire first-floor sector $s \in \{0, 1\}$. A scale-free true-interface argument, completed by a directed-rounding moving-boundary certificate, proves the complementary cone $s \geq 2$ throughout $K - \mu \geq \pi$. Combined with the no-mode theorem below the active wall, this closes the whole ordinary first-floor branch needed for the spectral theorem. In the no-drop branch we prove every $n = 1$ case and reduce the remaining data to finitely many discrete chambers, with the explicit bound

$n < 380$ in the two low-floor rows. Only the residual no-drop and high-floor compact walls remain uncertified; no assertion in this manuscript treats the full residual terminal-shelf scalar as proved.

1 Statement and status

For $0 < \rho < 1$ let

$$A^{(d)}_{\rho,1} = \{x \in \mathbb{R}^d : \rho < |x| < 1\}, \qquad d \geq 3,$$

and use the strict counting convention

$$N_D(A^{(d)}_{\rho,1}, K^2) := \#\{j : \lambda^D_j(A^{(d)}_{\rho,1}) < K^2\}.$$

Write

$$\omega_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}, \qquad L_d = \frac{\omega_d}{(2\pi)^d}.$$

The desired estimate is

$$N_D(A^{(d)}_{\rho,1}, K^2) \leq L_d |A^{(d)}_{\rho,1}| K^d = \frac{1-\rho^d}{2^d \Gamma(d/2+1)^2} K^d. \tag{1}$$

For $a > 0$, define the zero-extended ball action

$$G_a(z) = \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos \dfrac{z}{a}\right), & 0 \leq z \leq a, \\ 0, & z > a, \end{cases} \tag{2}$$

put $\mu = \rho K$, and set

$$A(z) = A_{\rho,K}(z) := G_K(z) - G_\mu(z). \tag{3}$$

For $x > 0$, let

$$[x]_< := \#\{n \in \mathbb{N} : n < x\} = \max\{0, \lceil x \rceil - 1\}.$$

In particular, $[n]_< = n - 1$ at a positive integral wall.

The only unproved statement on the primary path is the following.

Conjecture 1.1 (Positive shifted shell tails)**.** *For $0 < \rho < 1$, $K > 0$, and*

$$r \in \tfrac{1}{2}\mathbb{N},$$

one has

$$T_A(r) := [A(r) + \tfrac{1}{4}]_< + 2\sum_{j \geq 1} [A(r+j) + \tfrac{1}{4}]_< \leq 2 \int_r^K A(z)\,\mathrm{d}z. \tag{4}$$

The zero extension makes the sum finite and makes the assertion trivial when $r \geq K$.

Sections 2–5 prove that Conjecture 1.1 implies (1) in every dimension $d \geq 3$. Section 6 records the proved tail regions, the completed first interface-cell theorem, fixed-ratio and thin high-energy theorems, the terminal reserve, and the current finite reductions. It also states explicitly which finite wall problems remain.

2 General-dimensional spectral proxy

Put

$$a_d = \frac{d-2}{2}, \qquad \nu_{\ell,d} = \ell + a_d.$$

The degree-ℓ spherical harmonics on $\mathbb{S}^{d-1}$ have multiplicity

$$\kappa_{d,\ell} = \binom{\ell+d-1}{d-1} - \binom{\ell+d-3}{d-1} = \frac{(2\ell+d-2)(\ell+d-3)!}{\ell!(d-2)!}. \tag{5}$$

Here and below a binomial coefficient with lower index larger than its nonnegative upper index is zero.

Under the unitary radial substitution

$$u(r) = r^{(d-1)/2} R(r),$$

the degree-ℓ block becomes

$$L_{\ell,d} = -\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{\nu_{\ell,d}^2 - \frac{1}{4}}{r^2} \quad \text{on } (\rho, 1), \qquad u(\rho) = u(1) = 0. \tag{6}$$

Its positive eigenfrequencies are the zeros of

$$D_{\nu,\rho}(k) = J_\nu(\rho k) Y_\nu(k) - Y_\nu(\rho k) J_\nu(k), \qquad \nu = \nu_{\ell,d}.$$

With the standard continuous Bessel phase, let

$$\gamma_{\rho,K}(\nu) = \frac{\theta_\nu(K) - \theta_\nu(\rho K)}{\pi}.$$

The determinant factorization and phase monotonicity give the exact strict count

$$N_D(A_{\rho,1}^{(d)}, K^2) = \sum_{\ell \ge 0} \kappa_{d,\ell} [\gamma_{\rho,K}(\nu_{\ell,d})]_<. \tag{7}$$

The real-order phase-difference enclosure of [2, 3] gives, for $0 \le \nu \le K$,

$$\gamma_{\rho,K}(\nu) < A_{\rho,K}(\nu) + \frac{1}{4}. \tag{8}$$

For $\nu \ge K$, both the exact count and the action proxy vanish. Since $x < y$ implies $[x]_< \le [y]_<$, including when y is integral, (7)–(8) imply the wall-safe proxy bound

$$N_D(A_{\rho,1}^{(d)}, K^2) \le P_d^<(\rho, K), \qquad P_d^< := \sum_{\ell \ge 0} \kappa_{d,\ell} q_\ell, \tag{9}$$

where

$$q_\ell = [A(\ell + a_d) + \frac{1}{4}]_<. \tag{10}$$

The argument above is dimension-independent after the substitutions $\nu = \ell + a_d$ and (5). In particular, the published phase enclosure is already a real-order result: no interpolation is introduced when d changes parity.

3 Exact harmonic branching

Let $p = d - 2$, so $a_d = p/2$, and define

$$r_m = m + \frac{p}{2}, \qquad c_{d,m} = \binom{m+p-1}{p-1} = \binom{m+d-3}{d-3}. \tag{11}$$

Proposition 3.1 (Exact shifted-tail decomposition)**.** *For every finitely supported sequence $(q_\ell)_{\ell \geq 0}$,*

$$\sum_{\ell \geq 0} \kappa_{d,\ell} q_\ell = \sum_{m \geq 0} c_{d,m} \left(q_m + 2 \sum_{j \geq 1} q_{m+j} \right). \tag{12}$$

Consequently, for (10),

$$\boxed{P_d^< = \sum_{m \geq 0} c_{d,m} T_A(r_m).} \tag{13}$$

This identity is exact at every strict action wall.

Proof. The generating functions are

$$\sum_{\ell \geq 0} \kappa_{d,\ell} s^\ell = \frac{1+s}{(1-s)^{d-1}}, \qquad \sum_{m \geq 0} c_{d,m} s^m = \frac{1}{(1-s)^{d-2}},$$

and

$$1 + 2 \sum_{j \geq 1} s^j = \frac{1+s}{1-s}.$$

Coefficient comparison gives

$$\kappa_{d,\ell} = c_{d,\ell} + 2 \sum_{m=0}^{\ell-1} c_{d,m}.$$

Substitution and finite reindexing prove (12). For (10), one has $q_{m+j} = [A(r_m + j) + 1/4]_<$, which proves (13). No floor has been approximated or moved. $\square$

The shifts are

$$r_m \in \begin{cases} \mathbb{N}_0 + \frac{1}{2}, & d \text{ odd}, \\ \mathbb{N}, & d \text{ even}. \end{cases}$$

Thus Conjecture 1.1 is asked only on the two grids that actually occur.

4 The signed Weyl primitive

Define the right-continuous branching staircase

$$S_d(z) = \sum_{\substack{m \geq 0 \\ r_m \leq z}} c_{d,m} \tag{14}$$

and its discrepancy primitive

$$\Delta_d(z) = \int_0^z \left(S_d(u) - \frac{u^p}{p!} \right) \mathrm{d}u. \tag{15}$$

Proposition 4.1 (Signed branching primitive)**.** *For every $d \geq 3$ and $z \geq 0$,*

$$\boxed{\Delta_d(z) \leq 0.} \tag{16}$$

For $d = 3$, equality at positive z occurs precisely at the positive integers. For $d \geq 4$, the inequality is strict for $z > 0$.

Proof. For $0 \leq z < r_0 = p/2$, $S_d(z) = 0$, so

$$\Delta_d(z) = -\frac{z^{p+1}}{(p+1)!} \leq 0.$$

Let $M \geq 0$ and $r_M = M + p/2$. The hockey-stick identity gives

$$\int_0^{r_M} S_d(u)\,\mathrm{d}u = \sum_{m=0}^{M-1} (r_M - r_m) c_{d,m} = \binom{M+p}{p+1}.$$

Therefore

$$(p+1)!\Delta_d(r_M) = \prod_{j=0}^{p} (M+j) - \left(M + \frac{p}{2}\right)^{p+1} \leq 0, \tag{17}$$

because the arithmetic mean of $M, M+1, \ldots, M+p$ is $M + p/2$.

On the half-open cell $[r_M, r_{M+1})$,

$$S_d(z) = \binom{M+p}{p}, \qquad \Delta_d'(z) = \binom{M+p}{p} - \frac{z^p}{p!}.$$

If the critical point lies in this cell, it is

$$z_M^* = \left(\prod_{j=1}^{p} (M+j)\right)^{1/p}.$$

Direct substitution into the cell formula gives

$$(p+1)!\Delta_d(z_M^*) = p(z_M^*)^p \left(z_M^* - M - \frac{p+1}{2}\right) \leq 0, \tag{18}$$

since z_M^* is the geometric mean of $M+1, \ldots, M+p$ and their arithmetic mean is $M + (p+1)/2$. If the critical point is outside the cell, the cell maximum is at an endpoint, already covered by (17). This proves (16).

For $p = 1$, the cell maximum is $z = M + 1$ and has value zero. For $p \geq 2$, the consecutive factors in both AM–GM applications are unequal, which gives the stated strictness. □

5 Exact defect identity and conditional closure

Define

$$D_A(r) = 2\int_r^K A(z)\,\mathrm{d}z - T_A(r) \tag{19}$$

and the continuous payment

$$W_d = \frac{2}{p!} \int_0^K z^p A(z)\,\mathrm{d}z. \tag{20}$$

Lemma 5.1 (Exact Weyl normalization)**.** *For $p = d - 2$,*

$$\int_0^a z^p G_a(z)\,\mathrm{d}z = \frac{\Gamma((p+1)/2)}{4\sqrt{\pi}\,(p+2)\Gamma((p+4)/2)} a^{p+2}. \tag{21}$$

Consequently,

$$\boxed{W_d = \frac{1-\rho^d}{2^d\Gamma(d/2+1)^2} K^d = L_d |A^{(d)}_{\rho,1}| K^d.} \tag{22}$$

Proof. Scale $z = ax$ in (2), integrate the term containing $\arccos x$ by parts, and evaluate the resulting beta integrals. This gives (21). Multiplying the difference of (21) at $a = K$ and $a = \rho K$ by $2/p!$, the duplication formula for Γ reduces the coefficient to

$$\frac{1}{2^{p+2}\Gamma((p+4)/2)^2} = \frac{1}{2^d\Gamma(d/2+1)^2}.$$

The last equality in (22) follows from $|A^{(d)}_{\rho,1}| = \omega_d(1-\rho^d)$. ☐

Proposition 5.2 (Branching-defect identity)**.** *Let*

$$\mathcal{B}_d(A) = 2\int_0^K (-A'(z))(-\Delta_d(z))\,\mathrm{d}z. \tag{23}$$

Then $\mathcal{B}_d(A) \geq 0$ and

$$\boxed{W_d - P_d^< = \mathcal{B}_d(A) + \sum_{m\geq 0} c_{d,m} D_A(r_m).} \tag{24}$$

Proof. The shell action is absolutely continuous, decreasing, and vanishes at K. Fubini gives

$$\sum_{m\geq 0} c_{d,m}\, 2\int_{r_m}^K A(z)\,\mathrm{d}z = 2\int_0^K S_d(z)A(z)\,\mathrm{d}z.$$

Since $\Delta_d' = S_d - z^p/p!$ almost everywhere and both endpoint terms vanish, integration by parts gives

$$2\int_0^K S_d A = W_d + 2\int_0^K A\Delta_d' = W_d - 2\int_0^K A'\Delta_d = W_d - \mathcal{B}_d(A).$$

Both A' and Δ_d are nonpositive, so $\mathcal{B}_d(A) \geq 0$. Subtract (13) and use (19). ☐

Theorem 5.3 (Conditional all-dimensional shell theorem)**.** *If Conjecture 1.1 holds, then (1) holds for every $d \geq 3$, $0 < \rho < 1$, and $K \geq 0$. It then also holds for every shell $A^{(d)}_{r,R} = \{x \in \mathbb{R}^d : r < |x| < R\}$ and every spectral parameter.*

Proof. The case $K = 0$ is trivial, so assume $K > 0$. Every r_m in (11) lies in $\frac{1}{2}\mathbb{N}$, so the conjecture gives $D_A(r_m) \geq 0$. Equations (9), (24), and (22) yield

$$N_D(A^{(d)}_{\rho,1}, K^2) \leq P_d^< \leq W_d = L_d|A^{(d)}_{\rho,1}|K^d.$$

For a shell with outer radius R, take $\rho = r/R$ and $K = R\sqrt{\Lambda}$ and use Dirichlet dilation. The non-strict counting form follows by applying the strict estimate at $\Lambda + \varepsilon$ and letting $\varepsilon \downarrow 0$. ☐

6 Proved tail regions and the first interface cell

The action derivative is

$$-A'(z) = \frac{1}{\pi}\begin{cases}\arccos\dfrac{z}{K} - \arccos\dfrac{z}{\mu}, & 0 < z < \mu,\\ \arccos\dfrac{z}{K}, & \mu \le z < K.\end{cases} \tag{25}$$

Thus A is decreasing and concave on $(0,\mu)$, decreasing and convex on (μ, K), and A, A' match at μ.

Proposition 6.1 (Outer and terminal tails)**.** *Conjecture 1.1 holds in either of the following cases:*

1. $r \ge \mu$;
2. $G_K(r) \le 3/4$.

Proof. If $r \ge \mu$, then $A = G_K$ on $[r, K]$; the claim is trivial when $r \ge K$. Otherwise set $B = \lceil K - r\rceil$ and zero-extend

$$g(s) = G_K(r+s) \quad (0 \le s \le K - r)$$

to the integer interval $[0, B]$. Since $G_K(K) = G'_K(K) = 0$, this extension is nonnegative, decreasing, convex, 1/2-Lipschitz, and vanishes at B. The convex shifted-floor theorem of [1] therefore bounds its ordinary-floor tail by $2\int_0^B g$. At every sample,

$$[g(j) + 1/4]_< \le \lfloor g(j) + 1/4 \rfloor.$$

The strict tail is consequently bounded by $2\int_0^B g = 2\int_r^K G_K$.

If $G_K(r) \le 3/4$, monotonicity and $A \le G_K$ imply $A(r+j) + 1/4 \le 1$ for every $j \ge 0$. Every strict bracket is therefore zero. Notice that equality at 1 is harmless precisely because the bracket is strict. □

We next sharpen the prescribed first milestone. For the ball action set

$$T_K(r) = [G_K(r) + \tfrac14]_< + 2\sum_{j\ge 1}[G_K(r+j) + \tfrac14]_<, \qquad D_K(r) = 2\int_r^K G_K(z)\,\mathrm{d}z - T_K(r). \tag{26}$$

The wall-safe translation in the preceding proof gives $D_K(r) \ge 0$.

Lemma 6.2 (A low-action 1/3-Lipschitz reserve)**.** *Let g be strictly decreasing and convex on $[0, b]$, with $g(b) = 0$ and $\mathrm{Lip}(g) \le 1/3$. If*

$$\frac34 < g(0) < 1,$$

then

$$2\int_0^b g(z)\,\mathrm{d}z - \left([g(0) + \tfrac14]_< + 2\sum_{j\ge 1}[g(j) + \tfrac14]_<\right) > \frac{11}{16}. \tag{27}$$

The function is understood to be zero-extended after b.

Proof. Let $h = g(0)$ and let $x \in (0, b)$ be defined by $g(x) = 3/4$. Since $g(j) < 1$, a sample contributes exactly one if and only if $j < x$. Therefore the bracketed count is

$$2\lceil x\rceil - 1.$$

The Lipschitz bound and $g(b) = 0$ give

$$g(z) \geq \max\{h - z/3, 0\}, \qquad 2\int_0^b g(z)\,\mathrm{d}z \geq 3h^2 > \frac{27}{16}.$$

If $x \leq 1$, the count is one and (27) follows.

Suppose $x > 1$ and put $a = (h - 3/4)/x$. Convexity implies that, for $z \geq x$,

$$g(z) \geq \max\left\{\frac{3}{4} - a(z - x), 0\right\}.$$

Hence

$$\int_0^b g(z)\,\mathrm{d}z \geq \frac{3}{4}x + \frac{(3/4)^2}{2a} = x\left(\frac{3}{4} + \frac{9}{32(h - 3/4)}\right) > \frac{15}{8}x.$$

Since $2\lceil x\rceil - 1 < 2x + 1$, the defect is larger than

$$\frac{15}{4}x - (2x + 1) = \frac{7}{4}x - 1 > \frac{3}{4} > \frac{11}{16}.$$

□

Proposition 6.3 (Exact one-interface reduction)**.** *Assume*

$$0 < \mu - r < 1, \qquad r \in \tfrac{1}{2}\mathbb{N}.$$

Define

$$n_0 = [G_K(r) + \tfrac{1}{4}]_< - [G_K(r) - G_\mu(r) + \tfrac{1}{4}]_< \in \mathbb{N}_0 \tag{28}$$

and

$$\delta_\mu(r) = \int_r^\mu G_\mu(z)\,\mathrm{d}z.$$

Then

$$\boxed{D_A(r) = D_K(r) + n_0 - 2\delta_\mu(r).} \tag{29}$$

Moreover,

$$2\delta_\mu(r) < \frac{4(\mu - r)^{5/2}}{15\sqrt{\mu}} < \frac{4\sqrt{2}}{15} < 1. \tag{30}$$

Consequently, the shifted-tail inequality holds if any of the following is true:

1. $n_0 \geq 1$;
2. $G_K(\max\{r, K/2\}) \geq 1$;
3. $r \geq K/2$.

The only remaining one-interface configurations satisfy

$$\boxed{r < \frac{K}{2}, \qquad K < \frac{1}{\omega_0}, \qquad n_0 = 0, \qquad \omega_0 := G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}.} \tag{31}$$

In particular,

$$r \in \left\{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4, \frac{9}{2}\right\}.$$

Proof. Since $r+j>\mu$ for every $j\geq 1$, all samples after the first agree with the ball samples. Thus $T_A(r)=T_K(r)-n_0$. Subtracting $2\int_r^{\mu} G_\mu$ from the ball integral proves (29).

The turning estimate

$$G_\mu(z)<\frac{(\mu-z)^{3/2}}{3\sqrt{\mu}}\qquad (0<z<\mu)$$

integrates to (30), using $\mu>r\geq 1/2$ and $\mu-r<1$. If $n_0\geq 1$, then $D_K(r)\geq 0$ and (29) is positive.

For $r<K$, use the same integer endpoint $B=\lceil K-r\rceil$ and zero-extended function $g(s)=G_K(r+s)$. Put

$$s_*=\max\{0,K/2-r\}.$$

Then g is $1/3$-Lipschitz on $[s_*,B]$ and $g(s_*)=G_K(\max\{r,K/2\})$. The refined ordinary-floor theorem of [3, Theorem 3.4], followed by $[\,\cdot\,]_<\leq\lfloor\cdot\rfloor$, gives

$$D_K(r)\geq\frac{1}{2}\left\lfloor G_K(\max\{r,K/2\})\right\rfloor. \tag{32}$$

For $r\geq K$ the same inequality is trivial. Hence the second condition and (30) prove the claim.

If $r\geq K/2$ and $G_K(r)<1$, the translated ball action is globally $1/3$-Lipschitz. Its tail is either zero or Lemma 6.2 applies. In the latter case $D_K(r)>11/16>4\sqrt{2}/15$, so (29) is again positive. This proves the third condition.

If none of the three conditions holds, then $r<K/2$ and $G_K(K/2)=\omega_0K<1$, giving (31). The elementary rational bounds

$$\frac{265}{153}<\sqrt{3}<\frac{26}{15},\qquad \frac{333}{106}<\pi<\frac{355}{113}$$

(the square-root bounds follow by squaring) give $1/10<\omega_0<1/9$. Consequently

$$\frac{9}{2}<\frac{1}{2\omega_0}<5.$$

Since $r<K/2<1/(2\omega_0)$ and $r\in\frac{1}{2}\mathbb{N}$, only the nine displayed shifts remain. □

Remark 6.4 (The residual first-interface family)*.* Proposition 6.3 replaces the only unresolved part of the first-interface problem by a bounded nine-shift family: $2r<K<1/\omega_0$, $r<\mu<\min\{r+1,K\}$, and $n_0=0$. The inequality to be checked there is exactly

$$D_K(r)\geq 2\delta_\mu(r).$$

Proposition 6.5 and Lemma 6.6 below close this family.

Proposition 6.5 (Finite ball-wall reduction)**.** *In the residual family of Proposition 6.3, put*

$$B=G_K(r)+\frac{1}{4},\qquad q=[B]_<,\qquad e=B-q\in(0,1],$$

and set

$$E_r(K)=D_K(r)-e.$$

Then

$$2\delta_\mu(r)<e. \tag{33}$$

Consequently, the residual one-interface inequality follows from $E_r(K)\geq 0$.

On every open K-interval containing no ball-sample wall, E_r is strictly increasing. At a wall of the first sample $G_K(r)+1/4$, the two unit drops in $D_K(r)$ and e cancel, so E_r is continuous. At a wall of a later sample $G_K(r+j)+1/4$, $j \geq 1$, E_r drops by two for each crossed level. It is therefore enough to check the right-hand limits at the finitely many later-sample walls, together with the first positive-count wall and any lower endpoint $K = 2r$ at which the count is already positive.

More explicitly, all walls that can occur have the form

$$G_K(r+j) = n - \frac{1}{4}, \qquad j \geq 0, \qquad n \in \{1,2,3\}, \qquad r+j < 10. \tag{34}$$

Proof. Let $h = G_\mu(r)$. If $q = 0$, strict increase of $G_a(r)$ in the radius a gives $h < G_K(r) < B = e$. If $q \geq 1$, write $B = q + e$ and use the strict bracket convention; then $[B-h]_< = [B]_<$ holds exactly when $h < e$. Thus, in both cases,

$$n_0 = 0 \quad \Longleftrightarrow \quad [B-h]_< = [B]_< \quad \Longleftrightarrow \quad h < e.$$

When $q \geq 1$, equality $h = e$ lands on the integral wall q and causes a strict-count drop. The function G_μ is strictly convex on $[r,\mu]$, with endpoint values h and 0. It lies strictly below its endpoint chord, and therefore

$$2\delta_\mu(r) < h(\mu - r) < e,$$

which proves (33). If $T_K(r) = 0$, then $T_A(r) = 0$ and the desired inequality is immediate. Otherwise it is enough, by (29), to prove $D_K(r) \geq e$.

Away from sample walls the integer-valued tail and q are constant. Writing $x = r/K$, differentiation of the ball action and its tail integral gives

$$\pi E_r'(K) = K\left(\arccos x - x\sqrt{1-x^2}\right) - \sqrt{1-x^2}, \tag{35}$$

and

$$\pi E_r''(K) = \arccos x + x\sqrt{1-x^2} - \frac{x^2}{K\sqrt{1-x^2}}. \tag{36}$$

Here $x < 1/2$ and $K > 1$. Hence

$$\pi E_r''(K) > \frac{\pi}{3} - \frac{1}{2\sqrt{3}} > 0.$$

For $r \geq 1$, evaluation at $K = 2r$ gives

$$\pi E_r'(2r) = \frac{2\pi r}{3} - \frac{\sqrt{3}}{2}(r+1) > 0.$$

For $r = 1/2$, positivity of the tail count implies $G_K(r) > 3/4$. Since G_K is $1/2$-Lipschitz and vanishes at K, this forces $K > 2$. Formula (35) is already positive at $(r,K) = (1/2, 2)$, since

$$\pi E_{1/2}'(2) = 2\arccos\frac{1}{4} - \frac{3\sqrt{15}}{8} > \frac{2\pi}{3} - \frac{3}{2} > 0.$$

Equation (36) then keeps it positive. Thus E_r is strictly increasing between walls.

The stated jump rules follow directly from the weights $1, 2, 2, \ldots$ in $T_K(r)$. At a first-sample wall, e equals 1 and its right-hand limit is 0, exactly matching the unit right-hand drop of D_K. Simultaneous later-sample crossings are combined at the same wall abscissa. Finally, $K < 1/\omega_0 < 10$ and $G_K(z) < K/\pi < 10/3$. Thus only $n = 1, 2, 3$ and samples $r + j < 10$ can occur, and

$$\partial_K G_K(z) = \frac{1}{\pi}\sqrt{1 - \frac{z^2}{K^2}} > 0 \qquad (K > z)$$

shows that every pair (j, n) produces at most one wall. This proves the finite reduction. $\square$

Lemma 6.6 (Exact residual wall certificate)**.** *In the residual family of Proposition 6.5, if $T_K(r) > 0$, then*

$$E_r(K) > 0. \tag{37}$$

More precisely, $E_r(K) > 3/20$ for $r \le 4$, while for $r = 9/2$ one has

$$E_{9/2}(K) > \frac{1253}{555}.$$

Proof. Put

$$H_r(K) = 2\int_r^K G_K(z)\,\mathrm{d}z - G_K(r) - \frac{1}{4}, \qquad M_r(K) = \sum_{j\ge 1}[G_K(r+j) + \frac{1}{4}]_<.$$

The first-sample bracket cancels from the definition of E_r, including at its strict walls, and hence

$$E_r(K) = H_r(K) - 2M_r(K). \tag{38}$$

For $x = r/K$, direct integration gives

$$H_r(K) = \frac{A}{2} - \frac{A\arcsin x + B\sqrt{1-x^2}}{\pi} - \frac{1}{4}, \quad A = \frac{K^2}{2} + r^2 + r, \quad B = K\left(\frac{3r}{2} + 1\right). \tag{39}$$

We record rational envelopes sufficient to check every wall. For $0 \le x \le 13/20$, let

$$a_m = \frac{\binom{2m}{m}}{4^m(2m+1)}, \qquad b_m = \frac{\binom{2m}{m}}{4^m(2m-1)},$$

and set

$$L(x) = \sum_{m=0}^{6} a_m x^{2m+1}, \qquad U(x) = L(x) + \frac{x^{15}}{1-x^2},$$
$$V(x) = 1 - \sum_{m=1}^{6} b_m x^{2m}, \quad W(x) = V(x) - \frac{x^{14}}{1-x^2}.$$

The power series, whose omitted positive coefficients are smaller than one, give

$$L \le \arcsin x \le U, \qquad W \le \sqrt{1-x^2} \le V.$$

Together with $333/106 < \pi < 355/113$, they imply

$$\begin{aligned} \underline{G}(K,z) = K\left[\frac{113}{355}(W + xL) - \frac{x}{2}\right] &\le G_K(z) \\ \le K\left[\frac{106}{333}(V + xU) - \frac{x}{2}\right] &= \overline{G}(K,z), \end{aligned} \tag{40}$$

where now $x = z/K$, and

$$\frac{A}{2} - \frac{106}{333}(AU + BV) - \frac{1}{4} \le H_r(K) \le \frac{A}{2} - \frac{113}{355}(AL + BW) - \frac{1}{4}. \tag{41}$$

Thus every comparison below is a rational-number comparison.

Let $k_n(z)$ denote the unique solution of $G_K(z) = n - 1/4$. Substitution in (40) gives the following pairwise disjoint brackets:

wall	bracket	wall	bracket
$k_1(3/2)$	$(22/5, 9/2)$	$k_1(2)$	$(5, 26/5)$
$k_1(5/2)$	$(57/10, 29/5)$	$k_1(3)$	$(63/10, 32/5)$
$k_1(7/2)$	$(69/10, 7)$	$k_1(4)$	$(15/2, 38/5)$
$k_2(3/2)$	$(77/10, 39/5)$	$k_1(9/2)$	$(81/10, 41/5)$
$k_2(2)$	$(83/10, 17/2)$	$k_1(5)$	$(87/10, 44/5)$
$k_2(5/2)$	$(9, 227/25)$		

(40a)

These are all later-sample walls. Indeed, the same rational bounds give

$$\frac{227}{25} < \frac{1}{\omega_0} < \frac{459}{50}$$

and, at $K = 459/50$,

$$\overline{G}(K, 11/2) < \frac{18}{25} < \frac{3}{4}, \quad \overline{G}(K, 3) < \frac{8}{5} < \frac{7}{4}, \quad \overline{G}(K, 3/2) < \frac{9}{4} < \frac{11}{4}.$$

Monotonicity in K and z excludes, respectively, every omitted level-3/4, level-7/4, and level-11/4 or higher wall.

At a right-hand wall limit, the value of M_r is the event number in the following exact order:

r	$(z, n; M_r)$
$\frac{1}{2}$	$(\frac{3}{2}, 1; 1), (\frac{5}{2}, 1; 2), (\frac{7}{2}, 1; 3), (\frac{3}{2}, 2; 4), (\frac{9}{2}, 1; 5), (\frac{5}{2}, 2; 6)$
1	$(2, 1; 1), (3, 1; 2), (4, 1; 3), (2, 2; 4), (5, 1; 5)$
$\frac{3}{2}$	$(\frac{5}{2}, 1; 1), (\frac{7}{2}, 1; 2), (\frac{9}{2}, 1; 3), (\frac{5}{2}, 2; 4)$
2	$(3, 1; 1), (4, 1; 2), (5, 1; 3)$
$\frac{5}{2}$	$(\frac{7}{2}, 1; 1), (\frac{9}{2}, 1; 2)$
3	$(4, 1; 1), (5, 1; 2)$
$\frac{7}{2}$	$(\frac{9}{2}, 1; 1)$
4	$(5, 1; 1)$
$\frac{9}{2}$	$\varnothing$.

Equations (39)–(41) then give the strict rational bounds

r	first-wall lower	first-wall upper	activation or $K = 2r$	other-wall lower
$\frac{1}{2}$	$> 3/20$	$< 8/25$	$> 11/25$	$> 13/20$
1	$> 17/100$	$< 1/2$	$> 14/25$	$> 7/10$
$\frac{3}{2}$	$> 9/25$	$< 53/100$	$> 69/100$	$> 37/50$
2	$> 2/5$	$< 29/50$	$> 37/50$	$> 39/50$
$\frac{5}{2}$	$> 23/50$	$< 63/100$	$> 22/25$	$> 81/100$
3	$> 13/25$	$< 69/100$	$> 47/50$	$> 17/20$
$\frac{7}{2}$	$> 29/50$	$< 19/25$	$> 109/100$	$--$
4	$> 33/50$	$< 83/100$	$> 163/100$	$--$.

For each $r \leq 4$, the two first-wall columns concern the right-hand value at $G_K(r+1) = 3/4$. Its displayed upper bound is smaller than both competing lower bounds, so this is the controlling wall. Formula (35) shows that H_r, and hence E_r between its later-sample walls, is increasing throughout the positive-count residual region. For $r \le 3$ the tail is zero before its first-sample activation; for $r = 7/2, 4$ the displayed $K = 2r$ value owns the lower endpoint. Finally, for $r = 9/2$ there is no later wall and

$$E_{9/2}(9) = \frac{43}{2} - \frac{279\sqrt{3}}{8\pi} > \frac{1253}{555}.$$

This proves (37). A dependency-free replay of all 628 rational comparisons is supplied in `scripts/general_d_finite_wall_fraction_verify.py`; it is an audit of, not a numerical premise for, the displayed estimates. □

Theorem 6.7 (One-interface shifted tails)**.** *Conjecture 1.1 holds whenever*

$$0 < \mu - r < 1.$$

It also holds at the endpoint $r = \mu$ by Proposition 6.1.

Proof. The three regions in Proposition 6.3 are already proved there. In its residual family, if $T_K(r) = 0$, then $T_A(r) = 0$ and the claim is immediate. Otherwise Lemma 6.6 and Proposition 6.5 give

$$D_K(r) = E_r(K) + e > e > 2\delta_\mu(r).$$

The exact identity (29) now gives $D_A(r) > 0$. □

Proposition 6.8 (Fixed-ratio high-energy tails)**.** *For $0 < \rho < 1$, put*

$$c_\rho = \frac{\arccos \rho}{\pi}, \qquad d_\rho = 1 - 2c_\rho = \frac{2 \arcsin \rho}{\pi},$$

$$\eta_\rho = G_1\left(\max\left\{\rho, \frac{1}{2}\right\}\right), \qquad a_\rho = \frac{2\pi\rho}{1-\rho}, \qquad C_* = \frac{1}{2} + \frac{8\sqrt{2}}{15},$$

and

$$K_0(\rho) = \frac{a_\rho + 2\eta_\rho C_* + \sqrt{a_\rho^2 + 4a_\rho\eta_\rho C_* + \eta_\rho^2}}{2\eta_\rho^2}. \tag{42}$$

If $K \ge K_0(\rho)$, then Conjecture 1.1 holds for every $r \in \frac{1}{2}\mathbb{N}$.

Proof. Only $r < \mu$ requires proof. Set

$$n = \lfloor \mu - r \rfloor, \qquad q = r + n, \qquad B = \lceil K - r \rceil.$$

Thus $q \le \mu < q + 1$. For integers $a < b$, write

$$\mathbf{T}(f; a, b) = \frac{1}{2}\lfloor f(a) \rfloor + \sum_{j=a+1}^{b-1} \lfloor f(j) \rfloor + \frac{1}{2}\lfloor f(b) \rfloor.$$

The strict brackets in T_A are bounded above by ordinary floors. Additivity at the split point and $A \le G_K$ therefore give, when $n \ge 1$,

$$\frac{T_A(r)}{2} \le \mathbf{T}\left(A(r + \cdot) + \frac{1}{4}; 0, n\right) + \mathbf{T}\left(G_K(r + \cdot) + \frac{1}{4}; n, B\right). \tag{43}$$

For $n = 0$, the first term is absent and the same inequality follows by direct pointwise majorization.

Assume first $n \geq 1$ and put

$$h(t) = A(r+t) + \frac{1}{4}, \qquad m_j = \lfloor h(j) \rfloor.$$

If the nonincreasing sequence $(m_j)_{j=0}^n$ leaves its first shelf, let p be its last index on that shelf; otherwise put $p = n$. On $[0, n]$, h is decreasing, concave, and c_ρ-Lipschitz. The quantitative concave trapezoidal-floor theorem [3, Theorem 1.4], with the basic concavity bound in the no-drop case, yields

$$\mathbf{T}(h; 0, n) \leq \int_0^n A(r+t)\,\mathrm{d}t + \frac{n}{4} - \frac{1 - c_\rho}{2}(n - p). \tag{44}$$

For $n = 0$, take $p = 0$ and read this as an absent identity.

Let $\sigma = -A'$. On $0 < z < \mu$,

$$\sigma'(z) = \frac{1}{\pi\sqrt{\mu^2 - z^2}} - \frac{1}{\pi\sqrt{K^2 - z^2}} \geq \kappa := \frac{1-\rho}{\pi\rho K}.$$

Indeed, after subtracting the value at $z = 0$, the remaining bracket is decreasing in its radius parameter. Since $\sigma(0) = 0$,

$$A(x) - A(y) = \int_x^y \sigma(z)\,\mathrm{d}z \geq \frac{\kappa}{2}(y^2 - x^2) \geq \frac{\kappa}{2}(y - x)^2.$$

The samples at r and $r + p$ lie on the same floor shelf, so their action drop is smaller than one. Keeping the r-term in the preceding estimate and using $r \geq 1/2$ gives

$$p < \sqrt{r^2 + a_\rho K} - r \leq \sqrt{a_\rho K + \frac{1}{4}} - \frac{1}{2}. \tag{45}$$

For the convex suffix, set

$$s = \max\left\{q, \frac{K}{2}\right\}, \qquad t_* = s - r.$$

Then $t_* \in [n, B]$, the translated ball action is $1/2$-Lipschitz on $[n, B]$ and $1/3$-Lipschitz on $[t_*, B]$, and

$$G_K(s) \geq K\eta_\rho.$$

The quantitative convex theorem [3, Theorem 3.4] gives

$$\mathbf{T}\left(G_K(r + \cdot) + \frac{1}{4}; n, B\right) \leq \int_n^B G_K(r+t)\,\mathrm{d}t - \frac{1}{4}\lfloor K\eta_\rho \rfloor. \tag{46}$$

The only mismatch between the two integrals on the right of (43) and the shell integral is

$$\delta = \int_q^\mu G_\mu(z)\,\mathrm{d}z.$$

The turning bound from [3, Lemma 6.2] and $0 \leq \mu - q < 1$, $\mu > 1/2$, give

$$0 \leq \delta \leq \frac{2(\mu - q)^{5/2}}{15\sqrt{\mu}} < \frac{2\sqrt{2}}{15}. \tag{47}$$

Combining (43), (44), and (46) gives

$$\frac{T_A(r)}{2} \leq \int_r^K A(z)\,\mathrm{d}z + \delta - \frac{M}{4}, \quad M = \lfloor K\eta_\rho \rfloor + d_\rho(n - p) - p. \tag{48}$$

By (45),

$$M > K\eta_\rho - \sqrt{a_\rho K + \frac{1}{4}} - \frac{1}{2}.$$

The definition (42) is exactly the positive-root condition

$$K\eta_\rho - \sqrt{a_\rho K + \frac{1}{4}} \geq C_*.$$

Thus $M > 8\sqrt{2}/15 > 4\delta$, and (48) proves the result. All inequalities use ordinary floors on the majorizing side, so strict action walls require no limiting argument. $\square$

The fixed-ratio threshold is deliberately uniform in the shift, but it is not sharp as $\rho \uparrow 1$. Retaining the location of the first shelf improves the thin-shell scale by two powers of $1 - \rho$.

Lemma 6.9 (Optimized thin first-shelf loss)**.** *Let*

$$\rho = 1 - \varepsilon, \qquad 0 < \varepsilon \leq \frac{1}{100}, \qquad K\varepsilon^2 \geq \frac{125}{8}.$$

Use n, p from the proof of Proposition 6.8, put $m = n - p$, and set

$$d = d_\rho, \qquad \mathcal{R} = p - dm.$$

If $\mathcal{R} > 0$, then

$$\mathcal{R} < \frac{361}{80\sqrt{\varepsilon}}. \tag{49}$$

Proof. This is a scale-free consequence of the exact local slope and the same-floor condition. Put

$$y = \sqrt{\varepsilon}, \quad \varkappa = K\varepsilon^2, \quad P = py, \quad \mathfrak{r} = \mathcal{R}y, \quad S = (p+m)y.$$

Elementary trigonometric bounds give

$$d > \frac{9}{10}, \qquad S = P + \frac{P - \mathfrak{r}}{d} < \frac{19}{9}P. \tag{50}$$

Write

$$U = \mu - r, \qquad t = \frac{r}{\mu}, \qquad w = \frac{U}{\mu} = 1 - t,$$

and define

$$\mathcal{Q} = 1 + \frac{y^2}{\varkappa} + \frac{yS}{\varkappa}, \qquad \widehat{q} = \frac{y^2}{\rho}(\mathcal{Q} - 1).$$

The unconditional first-shelf bound $p^2 < 2\pi\rho K/\varepsilon$ and (50) give

$$w < \frac{1 + p + m}{\rho K} = \frac{y^4 + y^3 S}{\varkappa\rho} = \widehat{q} < \frac{y^4}{\varkappa\rho} + \frac{19}{9}y\sqrt{\frac{2\pi}{\varkappa\rho}}.$$

Here the first term is at most $8/1237500 < 1/1000$, while the square of the factor following $19/9$ is at most $352/86625 < 1/225$. Hence

$$w < \widehat{q} < \frac{1}{1000} + \frac{19}{135} < \frac{1}{7}, \qquad t > 1 - \widehat{q} > \frac{6}{7}. \tag{51}$$

Using

$$-A'(z) = \frac{1}{\pi}\left(\arccos\frac{z}{K} - \arccos\frac{z}{\mu}\right)$$

in the same-floor inequality $A(r) - A(r+p) < 1$ gives, after scaling,

$$P^2 < \mathcal{H}(P, \mathfrak{r}) := \frac{2\pi^2 \mathcal{Q}}{(1-\widehat{q})^2}. \tag{52}$$

On the preceding domain, $\mathcal{H}$ decreases in $\mathfrak{r}$ and

$$\partial_P \mathcal{H} < \frac{673816}{1366875} < \frac{1}{2}.$$

Let $B_* = 361/80$. If $\mathfrak{r} \ge B_*$, then $P \ge \mathfrak{r}$ and $P^2 - \mathcal{H}(P, B_*)$ is increasing for $P \ge B_*$. Indeed, every intermediate $P' \in [B_*, P]$ retains the unconditional shelf ceiling and satisfies $S(P', B_*) < 19P'/9$, so (51) and the displayed derivative bound hold throughout the comparison interval. Its minimum over $y \le 1/10$ and $\varkappa \ge 125/8$ is at $P = B_*$, $y = 1/10$, and $\varkappa = 125/8$. There

$$\mathcal{Q} = \frac{12869}{12500}, \qquad \widehat{q} = \frac{41}{137500},$$

and, using $\pi < 22/7$,

$$B_*^2 - 2\left(\frac{22}{7}\right)^2 \frac{12869/12500}{(1-41/137500)^2} = \frac{72581986185449}{5925464687161600} > 0.$$

This contradicts (52) and proves (49). All displayed comparisons are rational once the elementary bound for π is inserted; the dependency-free script `computations/round9_verify_thin_plateau_constants.py` replays the finite arithmetic. □

Proposition 6.10 (Thin high-energy shifted tails)**.** *Let $\rho = 1 - \varepsilon$ with $0 < \varepsilon \le 1/100$. If*

$$K \ge \frac{125}{8\varepsilon^2}, \tag{53}$$

then Conjecture 1.1 holds for every $r \in \frac{1}{2}\mathbb{N}$.

Proof. The outer tails are covered by Proposition 6.1. For an inner tail, retain the notation of Proposition 6.8. Before its final estimate, (48) gives

$$\frac{T_A(r)}{2} \le \int_r^K A(z)\,\mathrm{d}z + \delta - \frac{M}{4}, \qquad M = \lfloor K G_1(\rho) \rfloor - \mathcal{R},$$

with $4\delta < 8\sqrt{2}/15$. If $\mathcal{R} \le 0$, the convex gain already gives $M > 1 > 4\delta$. If $\mathcal{R} > 0$, Lemma 6.9,

$$G_1(1-\varepsilon) \ge \frac{2\sqrt{2}}{3\pi}\varepsilon^{3/2}, \qquad \sqrt{2} > \frac{140}{99}, \qquad \sqrt{2} < \frac{99}{70}, \qquad \pi < \frac{1571}{500},$$

and $\lfloor x \rfloor > x - 1$ give

$$M > \frac{2829397}{3732696} > \frac{132}{175} > \frac{8\sqrt{2}}{15} > 4\delta.$$

Thus (48) proves the claim. The proof uses only a positive shift and a unit-spaced sampling grid; it is unchanged on the integer and half-integer grids, and every threshold equality is owned by the strict-floor majorant. □

Corollary 6.11 (A uniform small-hole sector)**.** *If*

$$0 < \rho < \omega_0, \qquad K(\omega_0 - \rho) \geq \frac{1}{2} + \frac{8\sqrt{2}}{15},$$

then Conjecture 1.1 holds for every $r \in \frac{1}{2}\mathbb{N}$.

Proof. For $r < \mu$, use the split (43), basic concavity on the head, and (46) with $s = K/2$. This gives

$$\frac{T_A(r)}{2} \leq \int_r^K A(z)\,\mathrm{d}z + \delta - \frac{\lfloor \omega_0 K \rfloor - n}{4}.$$

Since $r \geq 1/2$,

$$\lfloor \omega_0 K \rfloor - n > K(\omega_0 - \rho) - \frac{1}{2} \geq \frac{8\sqrt{2}}{15} > 4\delta.$$

The outer tails were proved in Proposition 6.1. □

7 Cell recurrence and the corrected global target

For $x_j = r + j$, set

$$Q_j = [A(x_j) + \frac{1}{4}]_<, \qquad D_j = 2\int_{x_j}^K A(z)\,\mathrm{d}z - \left(Q_j + 2\sum_{k>j} Q_k\right).$$

Then

$$D_j - D_{j+1} = 2\int_{x_j}^{x_j+1} A(z)\,\mathrm{d}z - Q_j - Q_{j+1}. \tag{54}$$

If

$$e_j = A(x_j) + \frac{1}{4} - Q_j \in (0, 1]$$

and

$$C_j = 2\int_{x_j}^{x_j+1} A(z)\,\mathrm{d}z - A(x_j) - A(x_j + 1),$$

then

$$D_j - D_{j+1} = C_j - \frac{1}{2} + e_j + e_{j+1}. \tag{55}$$

On a fully inner cell,

$$C_j = -\int_0^1 u(1-u)A''(x_j + u)\,\mathrm{d}u \geq 0.$$

Individual increments in (55) can nevertheless be negative, and a negative increment need not be paid by the next floor-drop cell. Thus the viable block formulation must retain the terminal convex reserve. If J is the first index with $x_J \geq \mu$, the exact target is

$$\boxed{D_0 = \sum_{j=0}^{J-1}(D_j - D_{j+1}) + D_J \geq 0,} \tag{56}$$

where $D_J \geq 0$ is the translated convex ball-tail defect. A proof of (56) needs a quantitative, not merely nonnegative, lower bound for D_J or an equivalent matched shelf reserve.

The many-cell floor dynamics can in fact be compressed to one initial shelf and one terminal one-interface tail. This is the form used below.

Proposition 7.1 (Wall-safe first-shelf reduction)**.** *Suppose $r < \mu$ and put*

$$n = \lfloor \mu - r \rfloor, \qquad q = r + n, \qquad d_\rho = \frac{2 \arcsin \rho}{\pi}.$$

For $0 \le j \le n$, let

$$F_j = \left\lfloor A(r+j) + \frac{1}{4} \right\rfloor .$$

If this sequence leaves its first shelf, let p be the last index on that shelf; otherwise put $p = n$. Define

$$R_p = 2 \int_r^{r+p} A(z)\,\mathrm{d}z - 2pF_0. \tag{57}$$

Then, at ordinary and strict floor walls alike,

$$\boxed{D_A(r) \ge D_A(q) + R_p + \frac{d_\rho}{2}(n-p).} \tag{58}$$

Moreover, if

$$\varepsilon_j = A(r+j) + \frac{1}{4} - F_j \in [0,1), \qquad \sigma = -A',$$

then

$$R_p = \mathcal{C}_p + p\left(\varepsilon_0 + \varepsilon_p - \frac{1}{2}\right), \tag{59}$$

$$\mathcal{C}_p := -\int_0^p u(p-u)A''(r+u)\,\mathrm{d}u \ge 0, \tag{60}$$

and

$$R_p \ge -\frac{p}{2} + 2\int_0^p u\sigma(r+u)\,\mathrm{d}u. \tag{61}$$

In particular, with

$$\kappa = \frac{1-\rho}{\pi\rho K}, \qquad a_\rho = \frac{2\pi\rho}{1-\rho},$$

one has

$$p < \sqrt{r^2 + a_\rho K} - r. \tag{62}$$

Proof. Put $h(t) = A(r+t) + 1/4$. At half-tail scale, the strict trapezoidal sum

$$\mathbf{T}_<(h;a,b) := \frac{1}{2}[h(a)]_< + \sum_{j=a+1}^{b-1} [h(j)]_< + \frac{1}{2}[h(b)]_<$$

on $[0,p]$ is at most pF_0. If $p < n$, the ordinary floor of h drops immediately after the left endpoint of $[p,n]$. The quantitative concave trapezoidal-floor theorem [3, Theorem 1.4] gives

$$\mathbf{T}_<(h;p,n) \le \int_p^n A(r+t)\,\mathrm{d}t - \frac{d_\rho}{4}(n-p).$$

If $p = n$, this block is absent. Additivity at p and n leaves exactly one half of the strict tail beginning at q, and hence

$$\frac{T_A(r)}{2} \le pF_0 + \int_{r+p}^q A(z)\,\mathrm{d}z - \frac{d_\rho}{4}(n-p) + \frac{T_A(q)}{2}.$$

Subtracting this from $\int_r^K A$ proves (58). At an ordinary wall the strict bracket is smaller than the ordinary floor used in the majorant, so no limiting argument is needed.

Since $F_0 = F_p$, the trapezoidal identity gives (59)–(60). Also $F_0 \le A(r+p)+1/4$, and therefore

$$R_p \ge -\frac{p}{2} + 2\int_0^p (A(r+t) - A(r+p))\,\mathrm{d}t,$$

which is (61) after reversing the order of integration. Finally, on the inner branch

$$\sigma'(z) = \frac{1}{\pi\sqrt{\mu^2 - z^2}} - \frac{1}{\pi\sqrt{K^2 - z^2}} \ge \kappa, \qquad \sigma(0) = 0.$$

Equality of the two endpoint floors implies $A(r) - A(r+p) < 1$. Consequently

$$1 > \int_r^{r+p} \sigma(z)\,\mathrm{d}z \ge \frac{\kappa}{2}((r+p)^2 - r^2),$$

which is exactly (62). $\square$

The proposition identifies the sharp scalar which remains. The point q is a one-interface start, so Theorem 6.7 gives $D_A(q) \ge 0$ and, more precisely,

$$D_A(q) = D_K(q) + \lambda_q - 2\int_q^\mu G_\mu(z)\,\mathrm{d}z, \qquad \lambda_q = [G_K(q) + \tfrac{1}{4}]_< - [A(q) + \tfrac{1}{4}]_<. \tag{63}$$

Thus every tail satisfying

$$d_\rho(n-p) \ge p \tag{64}$$

is already proved by (58) and $R_p \ge -p/2$. Any residual tail must satisfy the explicit interface-band restriction

$$n < \frac{1+d_\rho}{d_\rho}\left(\sqrt{r^2 + a_\rho K} - r\right), \tag{65}$$

and the sole remaining inequality is

$$\boxed{D_A(q) + R_p + \frac{d_\rho}{2}(n-p) \ge 0.} \tag{66}$$

Unlike a local bad-cell pairing, this scalar retains the terminal convex reserve forced by the counterexamples above.

Lemma 7.2 (Scale-free shelf curvature)**.** *In the notation of Proposition 7.1, suppose $p \ge 1$ and put*

$$\Delta = A(r) - A(r+p).$$

Then

$$\mathcal{C}_p \ge \frac{p^2}{3(2r+p)}\,\Delta. \tag{67}$$

Consequently, since $\Delta = \varepsilon_0 - \varepsilon_p$ on the first ordinary-floor shelf,

$$R_p \ge \frac{p^2}{3(2r+p)}(\varepsilon_0 - \varepsilon_p) + p\left(\varepsilon_0 + \varepsilon_p - \frac{1}{2}\right). \tag{68}$$

Moreover,

$$A(0) - A(r) \le \frac{r\Delta}{2p}. \tag{69}$$

Proof. On the inner branch $\sigma = -A'$ is nonnegative, increasing, and convex, with $\sigma(0) = 0$. Indeed,

$$\sigma'(z) = \frac{1}{\pi\sqrt{\mu^2 - z^2}} - \frac{1}{\pi\sqrt{K^2 - z^2}},$$

and differentiating once more preserves the sign. Use the positive hinge representation

$$\sigma(z) = az + \int_{[0,\infty)} (z-s)_+ \,\mathrm{d}\nu(s), \qquad a \ge 0, \quad \nu \ge 0.$$

Both sides of (67) are linear in σ. For the generator $\sigma(z) = z$ their ratio is

$$\frac{p^3/6}{p(r+p/2)} = \frac{p^2}{3(2r+p)}.$$

For a hinge with $s \le r$, the same ratio is

$$\frac{p^3/6}{p(r-s+p/2)} \ge \frac{p^2}{3(2r+p)}.$$

If $r < s < r+p$ and $L = r+p-s$, the ratio is $p - 2L/3 \ge p/3$, which is again at least the required constant. Hinges to the right contribute zero. This proves (67), and (68) follows from (59).

Finally, convexity and $\sigma(0) = 0$ imply that $\sigma(z)/z$ is nondecreasing. Hence

$$\int_0^r \sigma(z)\,\mathrm{d}z \le \frac{r}{2}\sigma(r), \qquad \Delta \ge p\sigma(r),$$

which proves (69). □

Lemma 7.3 (Fractional-top convex reserve)**.** *Let g be nonnegative, strictly decreasing, convex, and C^1 on $[0,L)$, with $g(L) = 0$, and extend it by zero. Set*

$$v = g(0), \qquad c_v = -g'(0) > 0,$$

and

$$D_g = 2\int_0^L g(s)\,\mathrm{d}s - \left([g(0)+\tfrac14]_< + 2\sum_{j\ge1}[g(j)+\tfrac14]_<\right), \qquad B = [g(0)+\tfrac14]_<.$$

For $1 \le k \le B$, let y_k be the unique point satisfying $g(y_k) = k - 1/4$, and put $c_k = -g'(y_k)$. Then

$$\boxed{D_g \ge \sum_{k=1}^{B}\left(\frac{1}{2c_k} - 1\right) + \frac{(v-B)_+^2}{c_v}.} \tag{70}$$

Proof. Let $y(t)$ be the decreasing inverse of g, extended by zero for $t \ge g(0)$. Convexity of g makes y convex; the zero extension remains convex because its right derivative is zero. Layer cake gives

$$2\int_0^L g = 2\int_0^{g(0)} y(t)\,\mathrm{d}t.$$

The level $k - 1/4$ contributes exactly $2\lceil y_k\rceil - 1 \le 2y_k + 1$ to the strict tail, including when y_k is integral. The tangent to y at $k - 1/4$ gives

$$\int_{k-1}^{k} y(t)\,\mathrm{d}t \ge y_k - \frac14 y'(k - \tfrac14) = y_k + \frac{1}{4c_k}.$$

These disjoint unit intervals cover $[0, B]$. If $v > B$, the supporting tangent at the top endpoint is

$$y(t) \geq \frac{v - t}{c_v} \qquad (B \leq t \leq v),$$

and therefore the unused top interval contributes

$$2 \int_B^v y(t)\,\mathrm{d}t \geq \frac{(v - B)^2}{c_v}.$$

If $v \leq B$ this interval is empty. Adding this contribution proves (70). At a top action wall the equality level is absent from B, exactly as required by the strict convention. □

Corollary 7.4 (Exact-angle terminal reserve)**.** *At the one-interface point q, put*

$$B = [G_K(q) + \tfrac{1}{4}]_<, \qquad Q = [A(q) + \tfrac{1}{4}]_<, \qquad v_q = G_K(q), \qquad \beta_q = \frac{\arccos(q/K)}{\pi}.$$

For $1 \leq k \leq B$, let $\theta_k \in (0, \pi/2)$ be the unique solution of

$$\frac{K}{\pi}(\sin\theta_k - \theta_k \cos\theta_k) = k - \frac{1}{4}.$$

Then

$$\boxed{D_A(q) \geq \max\left\{0,\ \frac{\pi}{2}\sum_{k=1}^{B}\frac{1}{\theta_k} - Q - 2\int_q^{\mu} G_\mu(z)\,\mathrm{d}z + \frac{(v_q - B)_+^2}{\beta_q}\right\}}. \tag{71}$$

Proof. Apply Lemma 7.3 to $g(s) = G_K(q + s)$. At the level $k - 1/4$ the ball slope is $c_k = \theta_k/\pi$, while the top slope is β_q, so

$$D_K(q) \geq \sum_{k=1}^{B}\left(\frac{\pi}{2\theta_k} - 1\right) + \frac{(v_q - B)_+^2}{\beta_q}.$$

Insert this in (63); the $-B$ from the sum cancels the $+B$ in $\lambda_q = B - Q$. The other entry in the maximum is the completed one-interface theorem. □

Corollary 7.5 (Zero-level terminal triangle)**.** *Suppose $q \leq \mu < q + 1$ and $B = [G_K(q) + 1/4]_< = 0$. Put*

$$v = G_K(\mu), \qquad c = \frac{\arccos\rho}{\pi}.$$

Then the shifted shell tail beginning at q has zero discrete count and

$$\boxed{D_A(q) = 2\int_q^K A(z)\,\mathrm{d}z \geq 2\int_\mu^K G_K(z)\,\mathrm{d}z \geq \frac{v^2}{c}.} \tag{72}$$

Consequently the scalar (66) is nonnegative whenever

$$\frac{v^2}{c} \geq \frac{p - d_\rho(n - p)}{2}. \tag{73}$$

Proof. The condition $B = 0$ is $G_K(q) \leq 3/4$. Since $A \leq G_K$ and both actions decrease, every strict bracket in the shell tail is zero, including at the equality wall. On $[\mu, K]$ one has $A = G_K$. Moreover G_K is convex there and lies above its tangent at μ,

$$G_K(z) \geq v - c(z - \mu).$$

The tangent reaches zero no later than K, because $G_K(K) = 0$; integrating its positive triangle gives $2\int_\mu^K G_K \geq v^2/c$. Finally use $R_p \geq -p/2$ in (58). □

7.1 A certified first-floor large ray

We record a computer-assisted result for the ordinary first-floor branch $F_0 = 1$, $p < n$. Put

$$x = r + p, \qquad y = x + 1, \qquad s = n - p - 1, \qquad q = x + s + 1,$$

so that $q \le \mu < q + 1$. At the first-shelf wall $A(x) = 3/4$, set

$$v = A(y), \qquad \lambda = \frac{K - \mu}{\pi\mu}, \qquad c = \frac{1}{\pi}\arccos\frac{\mu}{K}, \qquad S_z = \sqrt{\mu^2 - z^2},$$

and define the terminal envelope

$$\mathcal{E}(y, v) = 2\int_y^\mu \big[v - \lambda(S_y - S_z)\big]_+ \,\mathrm{d}z + \frac{(v - \lambda S_y)_+^2}{c}. \tag{74}$$

Before specializing to the wall, concavity on the first shelf and its first drop cell proves (66) whenever

$$p\big(A(r) + A(x)\big) + A(x) + A(y) + \mathcal{E}(y, A(y)) - (1 + 2p) \ge 0.$$

At $A(x) = 3/4$ this becomes the weakened wall certificate

$$\mathcal{W} := p\left(A(r) - \frac{5}{4}\right) + v - \frac{1}{4} + \mathcal{E}(y, v) \tag{75}$$

For fixed (μ, r, p), every term before subtracting the constant in the unspecialized certificate is strictly increasing in K. Indeed, the sampled actions increase, while v/λ and λ increase in the integral in (74); on the positive outer branch, $(v - \lambda S_y)/\arccos(\mu/K)$ also increases. Thus the minimum in a fixed first-floor chamber is its post-crossing value at the unique wall $A(x) = 3/4$.

Theorem 7.6 (First-floor large-ray certificate)**.** *On the implicit wall $A(x) = 3/4$, one has*

$$\boxed{\mathcal{W} > 0} \qquad \text{if} \qquad r \ge \frac{3}{2}, \quad s \in \{0, 1\}, \quad x \ge 100. \tag{76}$$

The assertion holds on the continuous relaxation of r, p and $\alpha = \mu - q \in [0, 1]$, and therefore on the required half-integral and integral chamber parameters.

Proof. Write

$$\delta = K - \mu, \qquad t = x^{-1/3}, \qquad \kappa = t\delta, \qquad P = tp, \qquad \beta = \mu - x \in [s + 1, s + 2].$$

Changing variables $R = x + u$ in the wall equation gives the exact identity

$$\frac{3\pi}{4} = \int_\beta^{\beta+\delta} \frac{\sqrt{u(2x + u)}}{x + u}\,\mathrm{d}u. \tag{77}$$

The elementary bounds

$$\sqrt{\frac{u}{2x}} \le \frac{\sqrt{u(2x + u)}}{x + u} \le \sqrt{\frac{2u}{x}}$$

on the wall interval, with a preliminary contradiction if that interval reaches $u = x$, give

$$\frac{22}{25} < \kappa < \frac{731}{250}. \tag{78}$$

Moreover, a possible failure must have $0 < A(r) - 3/4 < 1/2$. Comparing the radius derivatives at r and x then gives

$$A(r) - \frac{3}{4} \geq \frac{(22/25)P}{2\pi(583/500)\sqrt{(1083/500)(3571/1000 + P)}}.$$

At $P = 41$ the right side is strictly larger than $1/2$; hence every possible failure lies in $0 \leq P < 41$.

It remains to certify a compact four-variable box while retaining the implicit wall. For $0 \leq \xi \leq 1$ put $w_\xi = \beta t + \xi\kappa$ and define

$$\begin{aligned}
\Phi_x(\xi) &= \frac{\sqrt{w_\xi(2 + t^2 w_\xi)}}{1 + t^2 w_\xi}, \\
\Phi_r(\xi) &= \frac{\sqrt{(w_\xi + P)(2 + t^2(w_\xi - P))}}{1 + t^2 w_\xi}, \\
\Phi_y(\xi) &= \frac{\sqrt{((\beta - 1)t + \xi\kappa)(2 + t^2((\beta + 1)t + \xi\kappa))}}{1 + t^2(\beta t + \xi\kappa)}.
\end{aligned}$$

At $R = \mu + \xi\delta$ these are the exact scaled radius profiles for x, r, y. For a profile F, put

$$T_2[F] = \frac{F(0) + 2F(1/2) + F(1)}{4}, \qquad M_2[F] = \frac{F(1/4) + F(3/4)}{2}.$$

The radius profile is increasing and strictly concave, so two-panel trapezoid and midpoint quadrature enclose every wall root by

$$\frac{\kappa}{\pi} T_2[\Phi_x] \leq \frac{3}{4} \leq \frac{\kappa}{\pi} M_2[\Phi_x]. \tag{79}$$

The same lower quadrature gives

$$\underline{A}_r = \frac{\kappa}{\pi} T_2[\Phi_r], \qquad \underline{v} = \frac{\kappa}{\pi} T_2[\Phi_y].$$

Put $L = \beta - 1$ and

$$H = \frac{3}{4} - \frac{\kappa\sqrt{t}\sqrt{\beta(2 + \beta t^3)}}{\pi(1 + \beta t^3)}, \qquad \vartheta = \frac{\kappa}{1 + \beta t^3}, \qquad \mathcal{Q} = \frac{\sqrt{\vartheta(2 + t^2\vartheta)}}{\pi}.$$

The anchored slope envelope gives $v - \lambda S_y \geq H$. When $H > 0$, (74) and $c/t \leq \mathcal{Q}$ therefore yield the stable lower certificate

$$t\mathcal{W} \geq \mathcal{C}_* := P\left(\underline{A}_r - \frac{5}{4}\right) + t\left(\underline{v} - \frac{1}{4} + 2LH\right) + \frac{H^2}{\mathcal{Q}}. \tag{80}$$

This expression has a finite continuous limit at $t = 0$.

The directed-rounding subdivision covers

$$0 \leq t \leq \frac{53861}{250000}, \qquad s + 1 \leq \beta \leq s + 2, \qquad 0 \leq P \leq 41,$$

encloses every compatible κ through (79) and (78), and retains the exact domain condition

$$1 - Pt^2 - \frac{3}{2}t^3 \geq 0$$

equivalent to $r \geq 3/2$. Each terminal box is proved either outside this domain, automatic from $\underline{A}_r \geq 5/4$, or strictly positive from $H > 0$ and (80). At 384-bit Arb precision the two runs process, respectively,

$$17697 \quad (s=0), \qquad 25367 \quad (s=1)$$

boxes, to maximum depths 21 and 22. The identical certificate passes at 512 bits, and a 768-bit replay with tighter wall-root bisection also passes. The verifier is `scripts/general_d_first_floor_ray_verify.py`. All acceptance tests use directed interval signs; floating point only orders the subdivision queue. This proves (76). □

The stronger certificate needed on the compact part of the wall keeps the actual interface geometry instead of the relaxed terminal envelope. Put

$$q = x + s + 1, \qquad \alpha = \mu - q \in [0,1],$$

and, on $A(x) = 3/4$, write

$$a_r = A(r), \qquad u = A(q), \qquad h = G_K(\mu), \qquad c = \frac{1}{\pi}\arccos\frac{\mu}{K}.$$

Concavity of A on $[r, \mu]$ and the outer tangent triangle give

$$\boxed{\mathcal{C}_{\mathrm{ff}} := p\left(a_r + \frac{3}{4}\right) + (s+1)\left(\frac{3}{4} + u\right) + \alpha(u+h) + \frac{h^2}{c} - (1+2p) \leq D_A(r).} \tag{81}$$

The right-hand side is interpreted as the post-crossing value. At the wall itself the strict bracket can only lower the count.

Theorem 7.7 (Finite first-floor wall certificate)**.** *On the implicit wall $A(x) = 3/4$,*

$$\boxed{\mathcal{C}_{\mathrm{ff}} > 0} \qquad \text{if} \qquad r \geq \frac{3}{2}, \quad s \in \{0,1\}, \quad x < 100. \tag{82}$$

Consequently Theorem 7.6 and (82) close the complete sector $r \geq 3/2$, $s \in \{0,1\}$.

Proof. For fixed x, the wall has the radius-average form

$$\int_\mu^{K(\mu)} \sqrt{1 - \frac{x^2}{R^2}}\,\mathrm{d}R = \frac{3\pi}{4},$$

and hence

$$K'(\mu) = \frac{\sqrt{1 - x^2/\mu^2}}{\sqrt{1 - x^2/K^2}} \in (0,1).$$

Along this wall $A(r)$ is nonincreasing, $A(q)$ is increasing, and both h and c are decreasing. These signs follow by differentiating the radius average; for example

$$\frac{\mathrm{d}}{\mathrm{d}\mu}A(z) = \frac{\sqrt{1 - x^2/\mu^2}}{\pi}\left(\frac{\sqrt{1 - z^2/K^2}}{\sqrt{1 - x^2/K^2}} - \frac{\sqrt{1 - z^2/\mu^2}}{\sqrt{1 - x^2/\mu^2}}\right),$$

and the squared ratio in parentheses has derivative with the sign of $z^2 - x^2$. Also $K' < 1$ makes μ/K increase, while direct differentiation gives $h' < 0$.

Thus, on an exact rational panel $\alpha_0 \leq \alpha \leq \alpha_1$, strict rational wall brackets $K_i^- < K(q + \alpha_i) < K_i^+$ give a one-sided lower bound for every term in (81): evaluate a_r, h at the upper endpoint with

K_1^-, evaluate u at the lower endpoint with K_0^-, and bound c from above at the lower endpoint with K_0^+. The wall roots are isolated by exact rational bisection, and all transcendental signs are then decided with directed-rounding Arb arithmetic.

The exact enumeration is

$$r = \frac{m}{2}, \quad 3 \le m \le 199, \qquad p \ge 0, \quad r + p < 100, \qquad s \in \{0, 1\}.$$

It contains 9801 chambers for each s, hence 19602 in total. At both 384 and 512 bits the verifier accepts 19606 panels after processing 19610 panels and isolating 39208 endpoint roots. Only $(r, p, s) = (3/2, 4, 0), \dots, (3/2, 7, 0)$ require one bisection in α; no deeper split occurs. A 768-bit replay with 160 root bisections per endpoint gives the identical counts. The checked script is `scripts/general_d_first_floor_finite_verify.py`. Since every leaf has a strictly positive Arb lower endpoint, this proves (82). □

Theorem 7.8 (Complete small-start certificate)**.** *On the implicit wall $A(x) = 3/4$, one has*

$$\boxed{\mathcal{C}_{\mathrm{ff}} > 0} \qquad \text{if} \qquad r \in \left\{\frac{1}{2}, 1\right\}, \quad s \in \{0, 1\}, \quad p \ge 0. \tag{83}$$

Together with Theorems 7.6 and 7.7, this closes the whole ordinary first-floor sector $s \in \{0, 1\}$.

Proof. For $x < 100$, the correlated wall-panel construction in the proof of Theorem 7.7 applies without change. The exact enumeration

$$r \in \left\{\frac{1}{2}, 1\right\}, \qquad s \in \{0, 1\}, \qquad p \ge 0, \qquad r + p < 100$$

contains 398 chambers. Directed-rounding replays at 384 and 512 bits process 420 panels, accept 409 leaves, and isolate 807 endpoint roots, with maximum depth one. A 1024-bit replay with 200 root steps has identical counts. The verifier is `scripts/general_d_first_floor_small_starts_verify.py`.

It remains to treat $x \ge 100$. Put $\beta = \mu - x = s + 1 + \alpha \in [1, 3]$ and $\delta = K - \mu$. The wall equation is

$$\frac{3\pi}{4} = \int_{\beta}^{\beta+\delta} \frac{\sqrt{v(2x + v)}}{x + v}\, \mathrm{d}v. \tag{84}$$

It first implies $\delta < x - \beta$: otherwise the integration interval contains $[x/2, x]$, where the integrand is at least $\sqrt{5}/3$, already a contradiction at $x = 100$. Hence

$$\delta \le \left(\frac{9\pi\sqrt{2}}{8}\right)^{2/3} x^{1/3} < \frac{731}{250} x^{1/3}.$$

With $T_* = 53861/250000 > 100^{-1/3}$ this gives, exactly,

$$\frac{K}{x} < 1 + 3T_*^3 + \frac{731}{250} T_*^2 < \frac{6}{5}.$$

For $f_z(R) = \sqrt{1 - z^2/R^2}$, the ratio f_r/f_x decreases with R. Since $r \le 1$, $x \ge 100$, and $K < 6x/5$,

$$\left(\frac{f_r(R)}{f_x(R)}\right)^2 \ge \frac{K^2 - r^2}{K^2 - x^2} > \frac{36x^2 - 25}{11x^2} \ge \frac{14399}{4400} > \frac{25}{9}.$$

Integration over $R \in [\mu, K]$ therefore yields

$$A(r) > \frac{5}{3}A(x) = \frac{5}{4}.$$

Finally, if $v = A(x+1)$, the one-cell action loss is smaller than $\arccos(\mu/K)/\pi < 1/2$, so $v > 1/4$. The weaker wall certificate

$$p\left(A(r) - \frac{5}{4}\right) + v - \frac{1}{4} + \mathcal{E}$$

is thus strictly positive, since $\mathcal{E} \geq 0$. This proves the analytic ray and completes (83). $\square$

The last possible first-floor escape requires a separate limiting estimate, because no complete terminal level is present below the first action wall.

Lemma 7.9 (Uniform critical first-floor gap)**.** *Let*

$$H_\kappa(X) = \frac{c_0}{\sqrt{\kappa}}((\kappa + X)^{3/2} - X^{3/2}), \qquad w = H_\kappa(0) = c_0\kappa,$$

where $c_0 = 2\sqrt{2}/(3\pi)$. For $0 < w < 3/4$, let $H_\kappa(M) = 3/4$ and $H_\kappa(N) = 1$, and set

$$\mathcal{L}(w) = -2\int_M^N (1 - H_\kappa(X))\,\mathrm{d}X + \frac{M}{2}, \qquad \mathcal{Z}(w) = \frac{\pi}{\sqrt{2}}w^2.$$

Then

$$\boxed{\mathcal{L}(w) + \mathcal{Z}(w) > \frac{\pi}{8\sqrt{2}}.} \tag{85}$$

At $w = 3/4$ the same conclusion holds with the smaller wall-safe terminal value $\pi/(2\sqrt{2})$.

Proof. The function $1 - H_\kappa$ is convex between M and N, so its endpoint chord gives $\mathcal{L} \geq (3M - N)/4$. Write

$$X = \kappa u(t), \qquad u(t) = \frac{(1-t^2)^2}{4t^2}, \qquad \frac{H_\kappa(X)}{w} = \frac{3/t + t^3}{4}.$$

If ξ, η are the parameters at M, N, then

$$w = \frac{3\xi}{3 + \xi^4} = \frac{4\eta}{3 + \eta^4}.$$

Using that u decreases, the ranges $0 < w \leq 1/2$, $1/2 \leq w \leq 2/3$, and $2/3 \leq w < 3/4$ give, respectively,

$$\frac{\sqrt{2}}{\pi}(\mathcal{L} + \mathcal{Z}) \geq \frac{1}{8} + \frac{85469}{51518208}, \qquad \frac{1}{8} + \frac{41923}{1281424}, \qquad \frac{1}{8} + \frac{743}{4608}.$$

Here one uses $\xi \leq (259/243)w$ in the first range, $\xi \leq (283/256)w$ in the second, and $\eta \geq 1/2$ in the third. At the wall the one-sided strict convention can lower the normalized terminal payment by only $1/16$, leaving $1/8 + 455/4608$. This proves (85). $\square$

Theorem 7.10 (Global exhaustion of the ordinary first floor)**.** *In the universal active region $K > \pi/(1-\rho)$, every negative ordinary first-floor first-drop scalar*

$$F_0 = 1, \qquad p < n, \qquad D_A(q) + R_p + \frac{d_\rho}{2}(n-p) < 0$$

belongs to a relatively compact subset of the exact shell parameter space. Consequently the possible discrete labels (r, n, p, Q, B) are bounded. The case $p = 0$ is automatic, and the conclusion includes the whole cone $s = n - p - 1 \geq 2$.

Proof. Put $\tau = \sqrt{1-\rho}$, $a = K - \mu$, and $\kappa = \tau a$. The fixed-ratio, compact-optical, and thin high-energy results already proved above show that an escaping negative sequence must have

$$\tau \to 0, \qquad a \to \infty, \qquad \tau\kappa = (1-\rho)a < \frac{125}{8},$$

while the first-drop interface bound gives $c_0\kappa < 3/4$. The exact noncancelling turning profile is

$$\mathcal{H}_{\tau,\kappa}(X) = \frac{1}{\pi}\int_0^\kappa \frac{\sqrt{(\kappa + X - u)[2\kappa - (\kappa + X + u)\tau^2]}}{\kappa - u\tau^2}\,\mathrm{d}u. \tag{86}$$

On the complete possible negative head its ratio to $H_\kappa(X)$ tends to one uniformly.

After passage to a subsequence, $\kappa \to \kappa_0 \in [0, 3/(4c_0)]$. If $\kappa_0 > 0$, exact level convergence, the first-shelf reduction, and the wall-safe terminal estimate give

$$\liminf \tau\left(D_A(q) + R_p + \frac{d_\rho}{2}(n-p)\right) \geq \mathcal{Z}(c_0\kappa_0) + \mathcal{L}(c_0\kappa_0) > 0$$

by Lemma 7.9.

If $\kappa_0 = 0$, put $Y = \kappa X$. Uniformly on the physical level window,

$$\mathcal{H}_{\tau,\kappa}(Y/\kappa) \longrightarrow H_0(Y) := \frac{\sqrt{2}}{\pi}\sqrt{Y}.$$

The 3/4- and 1-levels converge to $Y_{3/4} = 9\pi^2/32$ and $Y_1 = \pi^2/2$. The exact rescaled prefix then satisfies

$$\begin{aligned}\liminf \kappa\tau\left(R_p + \frac{d_\rho}{2}(n-p)\right) &\geq -2\int_{Y_{3/4}}^{Y_1}(1 - H_0(Y))\,\mathrm{d}Y + \frac{Y_{3/4}}{2} \\ &= \frac{17\pi^2}{192} > 0.\end{aligned}$$

The nonnegative one-interface reserve is enough on this ray. Both possible limits contradict negativity, so no negative sequence escapes. The two shift grids have $r \geq 1/2$, and $r < \rho K$; the fixed-ratio bound therefore also prevents an escape through $\rho \downarrow 0$. Relative compactness follows. It bounds the discrete labels, although it does not assert finiteness of the connected components of the remaining real-analytic wall set. ☐

Remark 7.11 (First-floor sectors before the cone certificate). Theorems 7.6 and 7.7 close $r \geq 3/2$, $s \in \{0, 1\}$, while Theorem 7.8 closes the remaining two starts. Thus the whole sector $s \in \{0, 1\}$ is proved. Theorem 7.10 independently removes every noncompact part of the complementary cone $s \geq 2$; the next theorem proves that full cone directly in the universal active region.

Theorem 7.12 (Active first-floor cone certificate)**.** *On the true ordinary first-floor interface $A(x) = 3/4$, let*

$$x = r + p, \qquad q = x + s + 1, \qquad \alpha = \mu - q \in [0, 1], \qquad \beta = s + 1 + \alpha = \mu - x.$$

If

$$r \geq \frac{1}{2}, \qquad p \geq 0, \qquad s \geq 2, \qquad K - \mu \geq \pi, \tag{87}$$

then the true-interface certificate (81) satisfies

$$\boxed{\mathcal{C}_{\mathrm{ff}} > 0.} \tag{88}$$

Equivalently, (88) holds throughout the continuous cone $\beta \geq 3$ in the closed universal active region. Together with the no-mode region (281), this closes the whole ordinary first-floor branch needed for the general-dimensional spectral theorem. It does not assert the standalone shifted-tail conjecture below the active wall.

Proof. Write

$$a_r = A(r), \qquad u = A(q), \qquad h = G_K(\mu), \qquad c = \frac{1}{\pi}\arccos\frac{\mu}{K}.$$

Concavity of A on $[x, \mu]$ gives

$$\beta u \geq \alpha\frac{3}{4} + (s+1)h.$$

Compressing the two middle trapezoids in (81) therefore yields the wall-safe lower bound

$$\mathcal{C}_{\mathrm{ff}} \geq p\left(a_r - \frac{5}{4}\right) + \beta\left(\frac{3}{4} + h\right) + \frac{h^2}{c} - 1. \tag{89}$$

This retains the true interface and absorbs all dependence on s and α into $\beta \geq 3$.

Define the scale-free ball action and wall variables

$$g(z) = \frac{\sqrt{1-z^2} - z\arccos z}{\pi}, \qquad \rho = \frac{\mu}{K}, \quad \xi = \frac{x}{K}, \quad \eta = \frac{r}{K},$$

$$w = g(\xi) - \rho g(\xi/\rho), \qquad e = \rho - \xi, \qquad \epsilon = 1 - \rho, \qquad g_\rho = g(\rho).$$

The radius-integral formula

$$w = \frac{1}{\pi}\int_\rho^1 \sqrt{1 - \frac{\xi^2}{R^2}}\,\mathrm{d}R \tag{90}$$

and the wall equation give

$$Kw = \frac{3}{4}, \qquad \beta = \frac{3e}{4w}, \qquad p = \frac{3(\xi-\eta)}{4w}, \qquad h = \frac{3g_\rho}{4w}. \tag{91}$$

Thus the cone and active conditions are exactly

$$e \geq 4w, \qquad 3\epsilon \geq 4\pi w. \tag{92}$$

The ratio of the radius profiles at r and x decreases with the radius. Averaging on the wall gives

$$a_r \geq \frac{3}{4}Q, \qquad Q = \sqrt{\frac{1-\eta^2}{1-\xi^2}}.$$

Exact one-variable minimization gives

$$(\xi - \eta)(3Q - 5) \geq -2(1-\xi), \tag{93}$$

$$(\xi - \eta)(3Q - 5) \geq -(1-\xi) \quad (\xi \geq 9/10). \tag{94}$$

For completeness, put $D = (\xi - \eta)/(1-\xi)$. The global comparison is immediate for $D \leq 2/5$ and otherwise reduces, after squaring, to

$$9D^4 - 32D^3 + 24D^2 + 12D - 4 > 0.$$

For (94), the only nontrivial case has $Q < 5/3$, whence $Q^2 \ge 1 + 4D/5$; after $D \ge 1/5$ the comparison reduces to

$$\frac{36}{5}D^3 - 16D^2 + 10D - 1 > 0.$$

The first polynomial has exact positive Bernstein coefficients on

$$[2/5, 1/2],\ [1/2, 1],\ [1, 3/2],\ [3/2, 2],\ [2, 3],\ [3, 4],$$

and is positive term by term for $D \ge 4$. The second has the same certificate on

$$[1/5, 1/2],\ [1/2, 1],\ [1, 6/5],\ [6/5, 2],\ [2, 3],$$

with a positive termwise tail for $D \ge 3$.

Multiplying (89) by $16w^2/9$ and using (93) gives

$$\frac{16w^2}{9}\mathcal{C}_{\mathrm{ff}} \ge J_2, \qquad J_2 = w\left(\frac{e}{3} - \frac{2\epsilon}{3}\right) + eg_\rho + \frac{g_\rho^2}{c} - \frac{16}{9}w^2. \tag{95}$$

In the near-tip range $\xi \ge 9/10$, (94) improves this to

$$\frac{16w^2}{9}\mathcal{C}_{\mathrm{ff}} \ge J_1, \qquad J_1 = w\left(\frac{2e}{3} - \frac{\epsilon}{3}\right) + eg_\rho + \frac{g_\rho^2}{c} - \frac{16}{9}w^2. \tag{96}$$

Put $t = e/\epsilon$. At fixed ρ, $w(t)$ is increasing and concave, and $w(t)/t$ decreases. At the physical endpoint $\xi = 0$ this ratio is $\epsilon^2/(\pi\rho)$. If $\rho \le 11/20$, it already exceeds $\epsilon/4$, contradicting $e \ge 4w$. On $11/20 \le \rho \le 39/50$, a 4096-panel directed check at $t_0 = 3/\pi$ proves

$$w(t_0) > \frac{3\epsilon}{4\pi}.$$

For $t \le t_0$ the cone constraint fails, and for $t \ge t_0$ the active constraint fails. Hence every admissible point has

$$\rho > \frac{39}{50}, \qquad \frac{\xi}{\rho} \ge \frac{\sqrt{7}}{4} > \frac{33}{50}. \tag{97}$$

If $t \ge 4$, (95) and the active upper bound on w give directly

$$J_2 \ge \epsilon w\left(\frac{2}{3} - \frac{4}{3\pi}\right) > 0. \tag{98}$$

The cap $\rho > 49/50$, $0 \le t < 4$, is also analytic. With

$$\rho_0 = \frac{49}{50}, \quad G_0 = \frac{2\sqrt{2\rho_0}}{3\pi}, \quad H_0 = \frac{2\rho_0}{3}, \quad B_0 = \frac{2\sqrt{2}}{3\pi\sqrt{\rho_0}}, \quad L(t) = (1+t)^{3/2} - t^{3/2},$$

the radius integrals imply

$$g_\rho \ge G_0\epsilon^{3/2}, \qquad \frac{g_\rho}{c} \ge H_0\epsilon, \qquad w \le B_0 L(t)\epsilon^{3/2}. \tag{99}$$

Here $\xi > 9/10$, so J_1 applies. For $0 \le t \le 1/2$, use $(1-2t)L(t) \le 1$; for $1/2 \le t < 4$, discard the nonnegative linear w-term. The resulting normalized lower margins are, respectively,

$$\frac{J_1}{\epsilon^{5/2}} > 0.0421933580845, \qquad \frac{J_1}{\epsilon^{5/2}} > 0.108941570080. \tag{100}$$

It remains only the compact strip $39/50 \le \rho \le 49/50$, $0 \le t \le 4$, subject to (92). The interval calculation follows the moving boundary $e = 4w$ rather than enclosing it by an independent box. At fixed ρ,

$$w_e = \frac{\arccos\xi - \arccos(\xi/\rho)}{\pi} > 0, \qquad w_{ee} = \frac{1}{\pi\sqrt{1-\xi^2}} - \frac{1}{\pi\sqrt{\rho^2-\xi^2}} < 0.$$

Thus $F(e) = e - 4w(e)$ is strictly convex and $F(0) = -4g_\rho < 0$; when the admissible interval is nonempty it has one beta-entry root. Substitution of $w = e/4$ at that root removes interval dependency and gives

$$J_b = \frac{g_\rho^2}{c} + eg_\rho - \frac{\epsilon e}{6} - \frac{e^2}{36}. \tag{101}$$

Moreover, with $z = g_\rho/w$ and

$$X = \frac{1}{3} - \frac{2\epsilon}{3e} - \frac{32w}{9e},$$

concavity, $0 < w_e \le (w - g_\rho)/e$, and differentiation of J_2 give

$$\frac{J_2'}{w} \ge \frac{1}{3} + z + (1-z)\min(0, X). \tag{102}$$

The directed-rounding certificate proves that both (101) and (102) are strictly positive on every accepted panel, so positivity propagates from the moving beta boundary to the active endpoint.

The verifier is `scripts/general_d_first_floor_s_cone_verify.py`. Independent replays at 256, 384, and 512 bits have identical topology: 507 ρ-panels, 253 ρ-splits, 254 positive panels and no empty panel, with 2422 derivative boxes, maximum ρ-depth 12, and maximum t-depth 5. The smallest certified boundary lower endpoints are $5.57124699231 \times 10^{-9}$, $5.57124666798 \times 10^{-9}$, and $5.57124693441 \times 10^{-9}$ at the three precisions; the smallest normalized derivative lower endpoint is 0.0010973528439 in each replay. All sign decisions use exact rational arithmetic or directed Arb intervals.

The low-rho exclusion, (98), the analytic cap, and the moving-boundary certificate cover the complete domain (87), proving (88). Theorems 7.6–7.8 cover $s \in \{0, 1\}$, while the present theorem covers $s \ge 2$ on the active side. On the complementary side the spectral counting function vanishes by (281); that final observation is used only for the spectral theorem and not as a below-active shifted-tail assertion. ☐

Proposition 7.13 (Exact no-drop identities)**.** *Suppose $p = n \ge 1$, and write $F_0 = \cdots = F_n = f$. Put*

$$\varepsilon_q = A(q) + \frac{1}{4} - f \in [0, 1), \qquad \chi_q = \mathbf{1}_{\{A(q)+1/4=f\}}.$$

Then

$$D_A(r) = D_A(q) + R_n + \chi_q, \tag{103}$$

$$R_n = -\frac{n}{2} + 2n\varepsilon_q + 2\int_0^n u\,\sigma(r+u)\,\mathrm{d}u. \tag{104}$$

Thus the no-drop branch is automatic when $f = 0$, or $\varepsilon_q \ge 1/4$, or

$$2\int_0^n u\,\sigma(r+u)\,\mathrm{d}u \ge n\left(\frac{1}{2} - 2\varepsilon_q\right).$$

Away from the endpoint wall, the exact shifted-tail condition is the quantitative payment

$$D_A(q) \geq \frac{n}{2} - 2n\varepsilon_q - 2\int_0^n u\,\sigma(r+u)\,\mathrm{d}u > 0. \tag{105}$$

At the endpoint wall the right side is reduced by the favorable unit $\chi_q = 1$ in (103); proving the wall-safe scalar (66) directly asks for the stronger displayed payment.

Proof. Strict decrease shows that a lower ordinary-floor wall cannot occur at an inner sample before q without ending the shelf. Hence the strict brackets at $r, \dots, q$ equal f, except that the bracket at q is $f-1$ when $\chi_q = 1$. Therefore

$$T_A(r) - T_A(q) = 2nf - \chi_q,$$

which proves (103). Since $f = A(q) + 1/4 - \varepsilon_q$,

$$\begin{aligned} R_n &= 2\int_0^n (A(r+t) - A(q))\,\mathrm{d}t + 2n\left(\varepsilon_q - \frac{1}{4}\right) \\ &= 2\int_0^n \int_t^n \sigma(r+u)\,\mathrm{d}u\,\mathrm{d}t + 2n\left(\varepsilon_q - \frac{1}{4}\right), \end{aligned}$$

and reversing the order proves (104). The remaining statements follow from $D_A(q) \geq 0$. $\square$

7.2 High floors: an automatic thin sector and the critical ray

Assume that the first ordinary floor is $F_0 = f \geq 2$, and write

$$\varepsilon = 1 - \rho, \qquad a = \varepsilon K = K - \mu, \qquad h = f - \frac{1}{4}, \qquad m = n - p. \tag{106}$$

Since $A(0) = a/\pi > A(r) \geq h$, every active high-floor configuration satisfies

$$a > \pi h. \tag{107}$$

The reduced scalar is

$$\mathcal{S} = D_A(q) + R_p + \frac{d_\rho}{2} m.$$

For $b > 0$, put $B_b(y) = \pi^{-1}\sqrt{b^2 - y^2}$. The radius-average formula for the shell action gives, for $0 \leq y \leq a\rho$,

$$\frac{1}{2}B_a(y) + \frac{1}{2\rho}B_{a\rho}(y) \leq \mathcal{A}_{\varepsilon,a}(y) := A_{1-\varepsilon,a/\varepsilon}(y/\varepsilon) \leq B_a(y). \tag{108}$$

Lemma 7.14 (Quantitative product-shelf margin)**.** *Let $f \geq 3$, $h = f - 1/4$, $a > \pi h$, and $y_b = \sqrt{a^2 - \pi^2 h^2}$. If $0 \leq y_0 \leq s \leq y_b$, then*

$$\boxed{2\int_{y_0}^{s} B_a(y)\,\mathrm{d}y - 2f(s - y_0) + \frac{a-s}{2} \geq \frac{f^2}{4a}.} \tag{109}$$

Proof. For fixed y_0 the left side is concave in s, so it is enough to use $s = y_0$ and $s = y_b$. The first value is at least $(a - y_b)/2 \geq \pi^2 h^2/(4a) > f^2/(4a)$. At $s = y_b$, put $R = a/\pi$. Differentiation with respect to y_0 gives $2(f - B_a(y_0))$; hence the minimum occurs where $B_a(y_0) = f$ if $R \geq f$, and at $y_0 = 0$ if $h \leq R < f$.

In the first case set $u = \sqrt{R^2 - h^2}$ and $v = \sqrt{R^2 - f^2}$. The chord estimate gives

$$\frac{\mathcal{L}}{\pi} > \frac{2R - 3u + v}{4} \geq \frac{3h^2/2 - f^2}{4R} > \frac{f^2}{4\pi^2 R}.$$

The last comparison follows from $\pi^2 > 9$ and $112f^2 - 216f + 27 > 0$. In the second case write $R = h + t$, $0 \leq t < 1/4$. Rationalizing the same chord estimate gives

$$\frac{\mathcal{L}}{\pi} \geq \frac{P(h,t)}{6R}, \qquad P(h,t) = (h+t)^2 - 4(1-t)^2 t(2h+t) \geq h^2 - 2h - \frac{1}{4}.$$

Here $18(h^2 - 2h - 1/4) - 3f^2 = 15f^2 - 45f + 45/8 > 0$ for $f \geq 3$. Again $\mathcal{L}/\pi > f^2/(4\pi^2 R)$, which is (109). $\square$

Proposition 7.15 (An automatic finite-shell sector)**.** *In the high-floor notation above, suppose*

$$f \geq 3, \qquad \varepsilon a^4 \leq \frac{f^3}{144}. \tag{110}$$

Then

$$\boxed{\varepsilon\left(R_p + \frac{d_\rho}{2}m\right) \geq \frac{f^2}{8a} > 0.} \tag{111}$$

In particular, (66) *holds without using a terminal reserve.*

Proof. Put $y_0 = \varepsilon r$, $s = \varepsilon(r+p)$, and $y_q = \varepsilon q$. Since $A(r+p) \geq h$ and the upper half of (108) holds, $s \leq y_b$. Conditions (107)–(110) imply $\rho > 99/100$ and $y_b \leq a\rho$: indeed

$$1 - \rho^2 < 2\varepsilon \leq \frac{f^3}{72a^4} < \frac{\pi^2 h^2}{a^2}.$$

Thus the lower half of (108) applies on the whole shelf. With

$$E = \int_0^{y_b} \left(B_a - \rho^{-1} B_{a\rho}\right) \mathrm{d}y,$$

direct rationalization, followed by $h \geq 11f/12$, gives

$$E \leq \frac{a^3(\rho^{-2} - 1)}{3\pi^2 h} < \frac{\varepsilon a^3}{11f} \leq \frac{f^2}{1584a}. \tag{112}$$

Moreover $y_q \geq a\rho - \varepsilon$, and hence

$$d_\rho(y_q - s) - (a - s) \geq -(1 - d_\rho)a - \varepsilon(a+1). \tag{113}$$

Writing $\rho = \cos\theta$ gives $1 - d_\rho = 2\theta/\pi \leq \sqrt{2\varepsilon}$; condition (110) then yields

$$\frac{(1-d_\rho)a}{2} \leq \frac{f^2}{24a}, \qquad \frac{\varepsilon(a+1)}{2} \leq \frac{f^2}{24a}.$$

Finally, the exact scaled identity is

$$\varepsilon\left(R_p + \frac{d_\rho}{2}m\right) = 2\int_{y_0}^{s} \mathcal{A}_{\varepsilon,a}(y)\,\mathrm{d}y - 2f(s - y_0) + \frac{d_\rho}{2}(y_q - s).$$

Combining the last four displays with Lemma 7.14 gives

$$\varepsilon\left(R_p + \frac{d_\rho}{2}m\right) \geq \frac{f^2}{4a} - \frac{f^2}{1584a} - \frac{f^2}{12a} > \frac{f^2}{8a}.$$

Ordinary shelf walls and strict terminal walls only improve these bounds. $\square$

The complementary unbounded scaling is the grazing ray

$$\varepsilon \downarrow 0, \qquad K\varepsilon^{3/2} \longrightarrow \kappa \in (0,\infty), \qquad n\sqrt{\varepsilon} = O(1). \tag{114}$$

Set $c_0 = 2\sqrt{2}/(3\pi)$. At the coordinate $X = (\mu - z)\sqrt{\varepsilon}$, the ball formula gives, locally uniformly,

$$A(z) \longrightarrow H_\kappa(X) := \frac{c_0}{\sqrt{\kappa}}\left((\kappa+X)^{3/2} - X^{3/2}\right), \qquad H_\kappa(0) = w := c_0\kappa. \tag{115}$$

If $B = [w + 1/4]_<$, the exact-angle reserve gives the wall-safe limit

$$\liminf_{\varepsilon\downarrow 0} \sqrt{\varepsilon}\, D_A(q) \geq \mathcal{T}(w) := \frac{\pi}{2\sqrt{2}} w^{1/3} \sum_{k=1}^{B} (k - \tfrac{1}{4})^{-1/3}. \tag{116}$$

Indeed, at the level $\tau_k = k - 1/4$,

$$\sqrt{\varepsilon}\,\frac{\pi}{2\theta_k} \longrightarrow \frac{\pi}{2\sqrt{2}} w^{1/3} \tau_k^{-1/3},$$

whereas the finite Q-term and the interface cap vanish after multiplication by $\sqrt{\varepsilon}$. The fractional-top term in (71) is nonnegative and is not needed below. At a limiting action wall, the strict value of B is the smaller one-sided complete-level sum and is therefore the safe convention.

Theorem 7.16 (No negative leading obstruction on the critical ray)**.** *For every fixed limiting first-floor level $f \geq 2$, each no-drop or first-drop configuration on* (114) *has a nonnegative leading scalar after* (116) *is included. If its limiting shelf/head prefix is negative, the terminal reserve exceeds the magnitude of that prefix strictly.*

Proof. Write $X = \kappa u$ and $t = \sqrt{1+u} - \sqrt{u} \in (0,1]$. Direct algebra gives

$$\frac{H_\kappa(X)}{w} = F(t) := \frac{3/t + t^3}{4}, \qquad u = \frac{(1-t^2)^2}{4t^2}, \qquad H_\kappa'(X) = \frac{3c_0}{2} t. \tag{117}$$

Let M be the point $H_\kappa(M) = h = f - 1/4$ and $N_f \geq M$ the point $H_\kappa(N_f) = f$. Scaling the first-shelf reduction and minimizing over its starting point shows that the worst first-drop prefix is

$$\mathcal{L} = -2\int_M^{N_f} (f - H_\kappa(X))\,\mathrm{d}X + \frac{M}{2}. \tag{118}$$

Indeed, the first shelf gives the displayed integral, the remaining post-drop length gives $M/2$ because $d_\rho \to 1$, and moving the starting point beyond the f-level adds only nonnegative shelf area. Since $0 \leq f - H_\kappa \leq 1/4$ on this interval, $\mathcal{L} < 0$ forces $N_f > 2M$.

This last inequality forces

$$w > \frac{3}{4}, \qquad t_f > \frac{1}{4}, \tag{119}$$

where t_f is the parameter at N_f. To see this without an asymptotic estimate, use $3/(4t) \leq F(t) \leq 1/t$ and $h/f \geq 7/8$. If $w \leq 3/4$, then $t_f < 3/10$ and

$$\frac{u_f}{u_h} \leq \left(\frac{7}{6}\right)^2 \left(\frac{400}{351}\right)^2 < 2,$$

contrary to $N_f > 2M$. If $t_f \leq 1/4$, the same computation gives

$$\frac{u_f}{u_h} < \left(\frac{6}{5}\right)^2 \left(\frac{100}{91}\right)^2 < 2.$$

Thus (119) holds. Equations (117)–(118) now give

$$-\mathcal{L} < \frac{N_f - M}{2} < \frac{1}{3c_0} = \frac{\pi}{2\sqrt{2}}.$$

On the other hand, $B \geq 1$ and the first term of (116) gives

$$\mathcal{T}(w) \geq \frac{\pi}{2\sqrt{2}} w^{1/3} (3/4)^{-1/3} > \frac{\pi}{2\sqrt{2}}.$$

This strictly pays a negative first-drop prefix. In the no-drop branch $M = 0$ and $w \geq h$. If $w \geq f$ its prefix is already nonnegative; otherwise $F(t_f) = f/w \leq f/h \leq 8/7$, so $t_f > 1/4$ and the same comparison applies. □

Corollary 7.17 (Uniform critical no-drop gap for $f = 2, 3$)**.** *For a no-drop sequence on* (114) *with fixed* $f \in \{2, 3\}$,

$$\boxed{\liminf_{\varepsilon \downarrow 0} \sqrt{\varepsilon}\, \mathcal{S} \geq \gamma_0 := \frac{\pi}{2\sqrt{2}} \left[\left(\frac{7}{3} \right)^{1/3} - 1 \right] > \frac{1}{3}.} \tag{120}$$

Proof. No-drop gives $w \geq f - 1/4 \geq 7/4$. If the prefix is negative, Theorem 7.16 bounds its magnitude by $\pi/(2\sqrt{2})$, while the first complete terminal level is at least $\pi(7/3)^{1/3}/(2\sqrt{2})$. If the prefix is nonnegative the estimate is stronger. Finally, $\pi/(2\sqrt{2}) > 15/14$ and $(7/3)^{1/3} > 59/45$. □

Theorem 7.18 (Global exhaustion of high-floor first drops)**.** *The set of negative high-floor first-drop configurations*

$$F_0 = f \geq 2, \qquad p < n, \qquad D_A(q) + R_p + \frac{d_\rho}{2}(n - p) < 0$$

is relatively compact in the exact shell parameter space. In particular, the possible discrete labels (f, n, p, Q, B) *are bounded, although the bounds furnished by this argument are not numerical.*

Proof. Write $\epsilon = 1 - \rho$, $s = \sqrt{\epsilon}$, $a = K - \mu$, $\kappa = sa$, $m = n - p$, $x = r + p$, and $h = f - 1/4$. If $H(t) = A(\mu - t)$, the first drop gives

$$W := H(0) = G_K(\mu) < h, \qquad c_0 \kappa < h.$$

Let $H(M) = h$ and let T be the first point at which $H(T) = f$, or put $T = \mu$ if that level is absent. The shelf inequalities and the bound $0 \leq f - H \leq 1/4$ on its negative part give

$$R_p + \frac{d_\rho}{2} m \geq \frac{(1 + d_\rho)M - T - d_\rho}{2}. \tag{121}$$

Thus negativity forces $T > (1 + d_\rho)M - d_\rho$.

The required uniform interface control follows from the shell-increment elasticity inequality

$$\frac{H(t_2) - W}{H(t_1) - W} \geq \sqrt{\frac{t_2(2\mu - t_2)}{t_1(2\mu - t_1)}} \qquad (0 < t_1 < t_2 \leq \mu). \tag{122}$$

Indeed, for each $R \in [\mu, K]$ the quotient of

$$\sqrt{1 - \mu^2/R^2 + t(2\mu - t)/R^2} - \sqrt{1 - \mu^2/R^2}$$

by $\sqrt{t(2\mu - t)}$ is nondecreasing; integration in R proves (122). Applying it at M, T along a negative sequence with $\rho \to 1$ yields

$$f - W \le 1 + \frac{\sqrt{3}}{2} + o(1). \tag{123}$$

As in the first-floor exhaustion, the fixed-ratio, compact-optical, and thin high-energy theorems leave only

$$s \to 0, \qquad a \to \infty, \qquad s\kappa < \frac{125}{8}.$$

The turning estimate $W = c_0\kappa + o(1)$ and (123) then pin

$$f - 1 - \frac{\sqrt{3}}{2} - o(1) \le c_0\kappa \le f - \frac{1}{4} + o(1). \tag{124}$$

They also confine the dangerous level band to a bounded turning window.

If $f \to \infty$, the root-free exact-angle reserve contains $B \ge f - O(1)$ complete levels. Since $sK^{1/3} = \kappa^{1/3}$ and $\sum_{k \le B}(k - 1/4)^{-1/3} \gg B^{2/3}$, it gives $sD_A(q) \to +\infty$, whereas (121) and the bounded turning window give $s(R_p + d_\rho m/2) \ge -O(1)$. Hence an escaping negative sequence has bounded f.

Pass finally to a subsequence with fixed f and $\kappa \to \kappa_0 > 0$. Exact profile and level convergence give

$$\liminf s\left(R_p + \frac{d_\rho}{2} m\right) \ge \mathcal{L}_f := -2\int_{M_0}^{N_0} (f - H_{\kappa_0}(X))\,\mathrm{d}X + \frac{M_0}{2}.$$

The wall-safe complete-level reserve has limit

$$\mathcal{T}(w) = \frac{\pi}{2\sqrt{2}} w^{1/3} \sum_{k=1}^{[w+1/4]_<} (k - \tfrac{1}{4})^{-1/3}, \qquad w = c_0\kappa_0.$$

If $\mathcal{L}_f < 0$, the exact prefix geometry forces $N_0 > 2M_0$. The t-parameterization in (117) then gives, uniformly for $f \ge 2$,

$$-\mathcal{L}_f < \frac{3\pi}{8\sqrt{2}}, \qquad \mathcal{T}(w) > \frac{\pi}{2\sqrt{2}},$$

and hence

$$\mathcal{L}_f + \mathcal{T}(w) > \frac{\pi}{8\sqrt{2}}. \tag{125}$$

If $\mathcal{L}_f \ge 0$, the same comparison forces a complete terminal level, so the sum is again positive. This excludes the last escape. Relative compactness follows, and compactness bounds the discrete labels without asserting a finite number of analytic wall components. $\square$

Theorem 7.19 (Explicit high-floor compact reduction)**.** *In the high-floor first-drop branch, put*

$$\mathcal{S} = D_A(q) + R_p + \frac{d_\rho}{2}(n - p), \qquad \epsilon = 1 - \rho, \qquad s = \sqrt{\epsilon}.$$

If $f \ge 2$, $p < n$, and $0 < s \le 10^{-4}$, then

$$\boxed{\mathcal{S} \ge 0.} \tag{126}$$

More precisely, the contradictory assumption $\mathcal{S} < 0$ forces $s\mathcal{S} > 1/2$; no unconditional half-unit gap is asserted.

Consequently, if a high-floor first-drop scalar is negative, then, with

$$K_* = 156250000000000000, \qquad N_* = 156249999999999999, \qquad F_* = 52083333333333333,$$

its continuous parameters satisfy

$$\boxed{\frac{1}{312500000000000000} < \rho < \frac{99999999}{100000000}, \qquad \frac{21}{4} < K < K_*,} \tag{127}$$

and its discrete data satisfy the exact finite ranges

$$\boxed{\begin{array}{c} r \in \tfrac{1}{2}\mathbb{N}, \qquad \tfrac{1}{2} \le r < K_*, \qquad 2 \le n \le N_*, \qquad 1 \le p \le n-1, \\ 2 \le f \le F_*, \qquad 0 \le Q \le f-1, \qquad Q \le B \le F_*. \end{array}} \tag{128}$$

Here the strict terminal counts are

$$Q = [A(q) + \tfrac{1}{4}]_<, \qquad B = [G_K(q) + \tfrac{1}{4}]_<.$$

The residual continuous chambers also retain

$$\mu = \rho K, \qquad q = r + n \le \mu < q + 1, \qquad K - \mu > \frac{7\pi}{4}, \tag{129}$$

$$\left\lfloor A(r+j) + \frac{1}{4} \right\rfloor = f \quad (0 \le j \le p), \qquad A(r+p+1) < f - \frac{1}{4}. \tag{130}$$

These bounds make the integer data finite; they do not certify the remaining compact analytic walls.

Proof. We give the quantitative transfer because its contradiction quantifier is essential. Write

$$a = K - \mu, \qquad \kappa = sa = s^3 K, \qquad m = n - p, \qquad x = r + p, \qquad h = f - \frac{1}{4},$$

and set $\widehat{H}(t) = A(\mu - t)$. Let M be the unique point with $\widehat{H}(M) = h$ and let T be the point with $\widehat{H}(T) = f$, using $T = \mu$ provisionally if that level is absent. Under the standing assumption $\mathcal{S} < 0$, the completed one-interface theorem gives $D_A(q) \ge 0$, while the exact shelf geometry gives

$$T > (1 + d_\rho)M - d_\rho, \qquad d_\rho = 1 - 2c_\rho, \qquad c_\rho = \frac{\arccos \rho}{\pi} < s. \tag{131}$$

The last inequality follows from $\arccos(1 - s^2) = 2\arcsin(s/\sqrt{2}) < \pi s$.

Put $W = \widehat{H}(0) = G_K(\mu)$. If $M \le 100$, the slope bound $\widehat{H}' \le c_\rho$ gives $W > h - 1/100$. If $M > 100$, then

$$\frac{T}{M} > 1 + (1 - 2s)\frac{99}{100} \ge r_0 := 1 + \frac{4999}{5000}\frac{99}{100}.$$

The exact shell-increment elasticity estimate

$$\frac{\widehat{H}(T) - W}{\widehat{H}(M) - W} \ge \sqrt{\frac{T(2\mu - T)}{M(2\mu - M)}} \ge \frac{T/M}{\sqrt{2T/M - 1}}$$

and the exact rational inequality $r_0/\sqrt{2r_0 - 1} > 23/20$ imply

$$\frac{f - W}{f - W - 1/4} > \frac{23}{20}.$$

This remains valid under the provisional absent-level convention, because $f - W$ then strictly dominates $\widehat{H}(T) - W$. Thus in every case

$$\boxed{W > f - \frac{23}{12} \ge \frac{1}{12}.} \tag{132}$$

Introduce the exact and critical scaled profiles

$$\mathcal{H}_{s,\kappa}(X) = A\left(\mu - \frac{X}{s}\right), \qquad H_\kappa(X) = \frac{c_0}{\sqrt{\kappa}}((\kappa + X)^{3/2} - X^{3/2}), \qquad c_0 = \frac{2\sqrt{2}}{3\pi},$$

and $w = H_\kappa(0) = c_0\kappa$. The noncancelling integral formula gives

$$\sqrt{1-s^2}\, w \le W \le \frac{w}{1-s^2}. \tag{133}$$

Since (132) implies $W > f/24$, $c_0\pi = 2\sqrt{2}/3 < 1$, and $(1-s^2)/(24s) > 1$,

$$A(0) = \frac{w}{c_0\pi s} > \frac{(1-s^2)f}{24s} > f.$$

Thus the level T really exists.

For $X = \kappa u$ the exact shell-envelope comparison yields

$$\mathcal{H}_{s,\kappa}(X) \ge \frac{2}{3\sqrt{2}} H_\kappa(X). \tag{134}$$

Writing

$$t = \sqrt{1+u} - \sqrt{u}, \qquad F(t) = \frac{3/t + t^3}{4}, \qquad \frac{H_\kappa(\kappa u)}{w} = F(t),$$

equations (132)– (134) give, at $X_T = sT$,

$$F(t_T) < 54, \qquad t_T > \frac{1}{72}, \qquad u_T = \frac{(1-t_T^2)^2}{4t_T^2} < 1296. \tag{135}$$

On this entire window the ratio of the exact integrand to the critical integrand is

$$\frac{\sqrt{1-\mathfrak{a}s^2}}{1-\mathfrak{b}s^2}, \qquad 0 \le \mathfrak{a} < 649, \qquad 0 \le \mathfrak{b} \le 1.$$

Hence, by direct rational estimates at $s \le 10^{-4}$,

$$\frac{999}{1000} < \frac{\mathcal{H}_{s,\kappa}(X)}{H_\kappa(X)} < \frac{1001}{1000} \qquad (0 \le X \le X_T). \tag{136}$$

We next record the forced-slope step. If $M \le 100$, the small-M pinning and (136) give

$$F(t_T) < \frac{100}{87}\left(\frac{1000}{999}\right)^2 < \frac{6}{5} < F(1/3).$$

If $M > 100$, put $X_M = sM$ and $X_x = s(\mu - x)$. Since $D_A(q) \ge 0$, negativity forces $P := R_p + d_\rho m/2 < 0$. Concavity of the exact profile gives

$$sR_p \ge -\frac{X_T - X_x}{4}, \qquad sm = X_x - s\alpha \ge X_x - s, \qquad X_x \ge X_M,$$

and therefore $X_T/X_M > 5/2$. If $t_T \le 1/3$, then (136) and $h/f \ge 7/8$ imply

$$F(t_M) > \frac{6}{7}F(t_T) \ge \frac{27}{14} > F(2/5), \qquad \frac{t_M}{t_T} < \frac{8}{7}\frac{1001}{999}\frac{1891}{1875} < \frac{7}{6}.$$

The exact formula for $u(t)$ would then give

$$\frac{X_T}{X_M} < \left(\frac{7}{6}\right)^2 \left(\frac{25}{21}\right)^2 < 2,$$

a contradiction. Thus in both cases

$$\boxed{t_T > \frac{1}{3}.} \tag{137}$$

The exact derivative formula

$$\mathcal{H}'_{s,\kappa}(X) = \frac{1}{\pi s}\left[\arccos(1 - s^2(1+u)) - \arccos\left(1 - \frac{s^2 u}{1 - s^2}\right)\right]$$

together with

$$\sqrt{2y} \leq \arccos(1-y) \leq \frac{\sqrt{2y}}{\sqrt{1 - y/2}}$$

and (135)– (137) gives the exact bound

$$\mathcal{H}'_{s,\kappa}(X) > \frac{7}{50} \quad (X_M \leq X \leq X_T), \qquad X_T - X_M < \frac{25}{14}. \tag{138}$$

The rational and directed-rounding comparisons suppressed in this derivative transfer, as well as the other numerical comparisons in the present proof, are replayed by `scripts/general_d_high_floor_compact_verify.py`; independent runs at 512 and 1024 bits both pass. The preceding prefix estimate now yields

$$sP > -\frac{25}{56} - \frac{s}{2}. \tag{139}$$

Finally, (136) and (137) imply

$$f < \frac{1001}{1000} wF(t_T) < \frac{1001}{1000} w \frac{61}{27}, \qquad w > \frac{54000}{61061} > \left(\frac{19}{20}\right)^3.$$

Equations (133) and $s \leq 10^{-4}$ then give $W > 3/4$, so the strict terminal bracket contains the first ball level. If θ_1 is its exact angle, the exact terminal reserve is wall-safe and gives

$$D_A(q) \geq \frac{\pi}{2\theta_1} - 1 - 2\int_q^\mu G_\mu(z)\,\mathrm{d}z.$$

The cap is smaller than $2/5$, while the angle inequality $\sin\theta - \theta\cos\theta \geq 2\theta^3/(3\pi)$ gives

$$s\frac{\pi}{2\theta_1} \geq \left(\frac{\pi^2 w}{6\sqrt{2}}\right)^{1/3} > w^{1/3} > \frac{19}{20}.$$

Consequently

$$sD_A(q) > \frac{19}{20} - \frac{7s}{5}. \tag{140}$$

Adding (139) and (140) gives, still under the assumption $\mathcal{S} < 0$,

$$s\mathcal{S} > \frac{19}{20} - \frac{25}{56} - \frac{19}{100000} > \frac{1}{2},$$

which is impossible. This proves (126), including every ordinary-floor and strict-bracket wall.

It remains to derive the displayed box. For $\rho \le 99/100$, the fixed-ratio scalar theorem proved above, with

$$\eta_\rho > \frac{49}{165000}, \qquad a_\rho < 623, \qquad C_* < \frac{13}{10},$$

excludes negativity once

$$K \ge K_0(\rho) < \frac{624}{(49/165000)^2} < 7.1 \cdot 10^9.$$

For $\rho > 99/100$, the thin high-energy theorem excludes $K\epsilon^2 \ge 125/8$, while the result just proved forces $\epsilon > 10^{-8}$ for a negative candidate. Hence

$$K < \frac{125}{8\epsilon^2} < K_*.$$

The larger value K_* is therefore a global cutoff. Negativity excludes $p = 0$, so $p \ge 1$, $n \ge 2$, and $r < \mu < K$, which yields $\rho > 1/(2K_*)$ and $n \le N_*$. Moreover $A(r) \ge 7/4$ and $A(0) = (K-\mu)/\pi > A(r)$ give $K - \mu > 7\pi/4$ and $K > 21/4$. Finally, using $\pi > 3$,

$$f \le A(r) + \frac{1}{4} < \frac{K}{3} + \frac{1}{4}, \qquad B < G_K(0) + \frac{1}{4} < \frac{K}{3} + \frac{1}{4},$$

so $f, B \le F_*$. The shelf drop gives $Q \le f - 1$, and the remaining ranges and chamber conditions follow directly from their ordinary-floor and strict-bracket definitions. This proves (127)– (130). ☐

Theorem 7.20 (Global high-floor scalar exclusion)**.** *In the high-floor first-drop branch, suppose*

$$f = F_0 \ge 2, \qquad p < n, \qquad \rho > \frac{99}{100}.$$

Then the reduced scalar is strictly positive:

$$\boxed{\mathcal{S} = D_A(q) + R_p + \frac{d_\rho}{2}(n-p) > 0.} \tag{141}$$

There is no restriction on K, on the terminal counts, or on an ordinary- or strict-floor wall.

Consequently every negative high-floor first-drop candidate satisfies, with

$$K_7 := \frac{16988400000000}{2401}, \qquad N_7 := 7075551852, \qquad F_7 := 2358517284,$$

the improved continuous bounds

$$\boxed{\frac{2401}{33976800000000} < \rho \le \frac{99}{100}, \qquad \frac{21}{4} < K < K_7 < 7.1 \cdot 10^9,} \tag{142}$$

and the exact finite integer ranges

$$\boxed{\begin{array}{c} r \in \tfrac{1}{2}\mathbb{N}, \qquad 1 \le 2r \le 14151103706, \qquad 2 \le n \le N_7, \qquad 1 \le p \le n-1, \\ 2 \le f \le F_7, \qquad 0 \le Q \le f-1, \qquad Q \le B \le F_7. \end{array}} \tag{143}$$

The chamber still obeys

$$\mu = \rho K, \qquad q = r + n \le \mu < q + 1, \qquad K - \mu > \frac{7\pi}{4}, \tag{144}$$

$$\left\lfloor A(r+j) + \frac{1}{4} \right\rfloor = f \quad (0 \le j \le p), \qquad A(r+p+1) < f - \frac{1}{4}. \tag{145}$$

These are improved finite bounds, not a certificate of the residual $\rho \le 99/100$ analytic walls.

Proof. Put

$$c = \frac{\arccos \rho}{\pi}, \qquad d = d_\rho = 1 - 2c, \qquad \widehat{H}(t) = A(\mu - t), \qquad W = \widehat{H}(0) = G_K(\mu).$$

The profile $\widehat{H}$ is increasing and concave on $[0, \mu]$, with

$$0 \leq \widehat{H}'(t) \leq c. \tag{146}$$

Write

$$h = f - \frac{1}{4}, \qquad x = r + p, \qquad m = n - p, \qquad t_x = \mu - x.$$

The first shelf and its first drop give $W < h$ and determine a unique M by $\widehat{H}(M) = h$, with

$$M \leq t_x < M + 1. \tag{147}$$

If the level f exists, let $\widehat{H}(T) = f$ and put $\zeta = 0$. If it is absent, put $T = \mu$ and

$$\zeta = f - \widehat{H}(T) \in (0, 1/4].$$

Thus the proof keeps the present- and absent-level faces separate.

Set $P = R_p + dm/2$. Since $t_x = m + \alpha$ with $\alpha = \mu - q \in [0, 1)$, (147) gives $m \geq M - 1$. Changing variables in the exact shelf integral gives

$$R_p = 2 \int_{t_x}^{t_x + p} (\widehat{H}(t) - f)\, \mathrm{d}t \geq -2 \int_M^T (f - \widehat{H}(t))\, \mathrm{d}t.$$

The function $f - \widehat{H}$ is convex on $[M, T]$ and has endpoint heights $1/4$ and ζ. Its endpoint chord therefore proves the wall-safe prefix estimate

$$\boxed{P \geq -\left(\frac{1}{4} + \zeta\right)(T - M) + \frac{d}{2}(M - 1).} \tag{148}$$

Put $\delta = h - W > 0$. The ratio assumption implies the strict bounds

$$c < \frac{1}{20}, \qquad d > \frac{9}{10}. \tag{149}$$

Suppose first that $\delta \geq 1$. By (146),

$$M \geq \frac{\delta}{c} > 20. \tag{150}$$

The shell-increment elasticity (122), together with $V(t) = t(2\mu - t)$, gives, for $0 < t_1 < t_2 \leq \mu$,

$$\frac{\widehat{H}(t_2) - W}{\widehat{H}(t_1) - W} \geq \sqrt{\frac{V(t_2)}{V(t_1)}} \geq \frac{t_2/t_1}{\sqrt{2t_2/t_1 - 1}}. \tag{151}$$

If the level f exists, applying this at M, T yields

$$\frac{T/M}{\sqrt{2T/M - 1}} \leq 1 + \frac{1}{4\delta} \leq \frac{5}{4},$$

and hence $T/M \leq 5/2$. Equations (148) and (150) now give

$$P \geq \left(\frac{d}{2} - \frac{3}{8}\right) M - \frac{d}{2} > 1. \tag{152}$$

If the level is absent, put $a = 1/4 - \zeta \in [0, 1/4]$. Then (151) gives

$$\frac{\mu/M}{\sqrt{2\mu/M - 1}} \le 1 + \frac{a}{\delta} \le 1 + a.$$

For $R \ge 1$, the upper solution of $r/\sqrt{2r-1} = R$ is

$$r_*(R) = R^2 + R\sqrt{R^2 - 1}.$$

On $0 \le a \le 1/4$ one has the strict elementary estimate

$$\left(\frac{1}{2} - a\right)(r_*(1+a) - 1) < \frac{19}{50}. \tag{153}$$

Indeed, with $x = \sqrt{a}$ and

$$U(x) = \left(\frac{1}{2} - x^2\right)\left(2x^2 + x^4 + \frac{3}{2}x(1+x^2)\right),$$

the bound $\sqrt{a(2+a)} \le 3\sqrt{a}/2$ reduces the left side to $U(x)$. After $y = 2x$, all degree-32 Bernstein coefficients of $19/50 - U(y/2)$ are positive; the smallest is $309/396800$. Thus, on this absent-level face,

$$P > \left(\frac{d}{2} - \frac{19}{50}\right)M - \frac{d}{2} > \frac{9}{10} > 0. \tag{154}$$

This completes both level cases when $\delta \ge 1$.

It remains to consider $0 < \delta < 1$. The strict interface count is now fixed without a wall limit:

$$B_0 := [W + \tfrac{1}{4}]_< = f - 1, \qquad W = f - \frac{1}{4} - \delta > f - \frac{5}{4} \ge \frac{3}{4}. \tag{155}$$

Put

$$y = f - W = \delta + \frac{1}{4} \in \left(\frac{1}{4}, \frac{5}{4}\right).$$

We first prove the scale-free level-distance estimate

$$\boxed{T < \frac{3y}{c},} \tag{156}$$

which in particular proves that the level f is present.

For fixed ρ, the increment $E_K(t) = \widehat{H}(t) - W$ has the scaling form $E_K(t) = KE_1(t/K)$. Since E_1 is concave and $E_1(0) = 0$,

$$\partial_K E_K(t) = E_1(u) - uE_1'(u) \ge 0, \qquad u = t/K.$$

At fixed y, the distance to the level $W + y$ is therefore largest at the smallest admissible W. Since $W = f - y \ge 2 - y$, it suffices to take $f = 2$ and $W = 2 - y$.

Let

$$\theta = \arccos\rho, \qquad \epsilon = 1 - \rho, \qquad \Phi(\theta) = \sin\theta - \theta\cos\theta, \qquad W = K\frac{\Phi(\theta)}{\pi},$$

and set

$$t_0 = \frac{3y}{c} = \frac{3\pi y}{\theta}, \qquad u_0 = \frac{t_0}{\mu}.$$

The elementary trigonometric bounds valid for $\rho > 99/100$ are

$$\theta^2 < \frac{21}{1000}, \qquad \frac{499}{1000}\theta^2 < \epsilon \le \frac{1}{2}\theta^2, \qquad \frac{9979}{30000}\theta^3 < \Phi(\theta) < \frac{1}{3}\theta^3. \tag{157}$$

They imply

$$u_0 < \frac{9}{250}, \qquad \frac{u_0}{\epsilon} > \frac{199}{100}\frac{y}{W}. \tag{158}$$

In particular $t_0 < \mu$. Directly from the ball actions,

$$E_K(t_0) = \frac{\mu}{\pi}\int_0^{u_0} \left[\arccos(\rho(1-u)) - \arccos(1-u)\right] \, \mathrm{d}u. \tag{159}$$

Using

$$\sqrt{2z} \le \arccos(1-z) \le \frac{\sqrt{2z}}{\sqrt{1-z/2}},$$

then setting $u = \epsilon v$ and retaining the positive interval $0 \le v \le V$, where

$$V = \frac{199}{100}\frac{y}{W},$$

gives

$$E_K(t_0) > \frac{147}{100}W\int_0^V \left[\sqrt{1+\frac{99}{100}v} - \frac{101}{100}\sqrt{v}\right] \, \mathrm{d}v. \tag{160}$$

The integrand is positive on the retained interval. Define

$$I(V) = \frac{200}{297}\left(\left(1+\frac{99}{100}V\right)^{3/2} - 1\right) - \frac{101}{150}V^{3/2}.$$

With $x = y/W = y/(2-y)$, one has $1/7 < x < 5/3$. The function

$$J(x) = \frac{147}{100}I\left(\frac{199}{100}x\right) - x$$

is concave, because $I'' < 0$, and directed endpoint evaluation gives

$$J(1/7) > \frac{3}{20}, \qquad J(5/3) > \frac{7}{50}. \tag{161}$$

Thus (160) gives $E_K(t_0) > Wx = y$, proving (156).

Here $\zeta = 0$. Multiplying (148) by c, and using $cM \ge \delta$, $cT < 3(\delta + 1/4)$, (149), and $d < 1$, yields

$$cP > -\frac{3}{4}\left(\delta + \frac{1}{4}\right) + \frac{7}{10}\delta - \frac{1}{40} > -\frac{21}{80}. \tag{162}$$

At the terminal start put

$$B = [G_K(q) + \tfrac{1}{4}]_<, \qquad Q = [A(q) + \tfrac{1}{4}]_<.$$

Since $q \le \mu$ and the ordinary shelf has dropped by $x + 1 \le q$, the strict counts satisfy

$$B \ge B_0 = f - 1, \qquad Q \le f - 1. \tag{163}$$

For $1 \le k \le f-1$, the strict inequality in (155) gives

$$k - \frac{1}{4} \le f - \frac{5}{4} < W = G_K(\mu).$$

Hence every retained ball angle in Corollary 7.4 satisfies $\theta_k < \theta = \pi c$. The exact-angle reserve and the cap bound (30) therefore give

$$D_A(q) > \frac{f-1}{2c} - (f-1) - \frac{2}{5}. \tag{164}$$

After multiplication by c, the right side is minimized at $f = 2$; thus

$$cD_A(q) > \frac{1}{2} - \frac{1}{20}\frac{7}{5} = \frac{43}{100}. \tag{165}$$

Adding (162) and (165) proves the strict margin

$$\boxed{c\mathcal{S} > \frac{43}{100} - \frac{21}{80} = \frac{67}{400} > 0.} \tag{166}$$

Together with (152) and (154), this proves (141) on every wall.

For completeness, the finite algebraic and transcendental comparisons in (153) and (157)– (161) are replayed by `scripts/general_d_high_floor_global_scalar_verify.py`. Independent directed runs at 384, 768, and 1024 bits give the same strict margins.

It remains to derive the improved residual box. The fixed-ratio scalar estimate already used in the preceding compactness argument has the threshold (42). Uniformly for $\rho \le 99/100$, its audited constants

$$\eta_\rho > \frac{49}{165000}, \qquad a_\rho < 623, \qquad C_* < \frac{13}{10}, \qquad \eta_\rho < \frac{1}{3}$$

give

$$K_0(\rho) < \frac{624}{(49/165000)^2} = \frac{16988400000000}{2401} = K_7. \tag{167}$$

Thus a negative scalar has $\rho \le 99/100$ and $K < K_7$.

Negativity excludes $p = 0$, so $p \ge 1$ and $n \ge 2$. Since $r < \mu < K < K_7$ and $r \ge 1/2$,

$$\rho > \frac{1}{2K_7} = \frac{2401}{33976800000000}, \qquad 2r \le \lfloor 2K_7 \rfloor = 14151103706,$$

and, from $r + n \le \mu < K_7$,

$$n \le \left\lfloor K_7 - \frac{1}{2} \right\rfloor = N_7.$$

Moreover, $A(r) \ge 7/4$ and $A(0) > A(r)$ imply $K - \mu > 7\pi/4$ and $K > 21/4$. Finally, using $\pi > 3$,

$$f \le A(r) + \frac{1}{4} < \frac{K}{3} + \frac{1}{4}, \qquad B < G_K(0) + \frac{1}{4} < \frac{K}{3} + \frac{1}{4},$$

and exact integer division gives

$$\left\lfloor \frac{K_7}{3} + \frac{1}{4} \right\rfloor = 2358517284 = F_7.$$

The shelf drop gives $Q \le f - 1$, while $A \le G_K$ gives $Q \le B$. The remaining bounds and chamber conditions are their defining ordinary-floor and strict-bracket relations. This proves (142)–(145). $\square$

Theorem 7.21 (Extended global high-floor scalar exclusion)**.** *In the high-floor first-drop branch, suppose*

$$f = F_0 \geq 2, \qquad p < n, \qquad \frac{603}{625} \leq \rho < 1.$$

Then the reduced scalar is strictly positive:

$$\boxed{\mathcal{S} = D_A(q) + R_p + \frac{d_\rho}{2}(n-p) > 0.} \tag{168}$$

There is no restriction on K, on the terminal counts, or on an ordinary- or strict-floor wall.

Consequently every remaining negative high-floor first-drop candidate satisfies the continuous bounds

$$\boxed{\frac{1}{87889536} < \rho < \frac{603}{625}, \qquad \frac{21}{4} < K < 43944768,} \tag{169}$$

and the exact finite integer ranges

$$\boxed{\begin{gathered} r \in \tfrac{1}{2}\mathbb{N}, \qquad 1 \leq 2r \leq 87889535, \\ 2 \leq n \leq 43944767, \qquad 1 \leq p \leq n-1, \\ 2 \leq f \leq 14648256, \\ 0 \leq Q \leq f-1, \qquad Q \leq B \leq 14648256. \end{gathered}} \tag{170}$$

The chamber also retains

$$\mu = \rho K, \qquad q = r + n \leq \mu < q + 1, \qquad K - \mu > \frac{7\pi}{4}, \tag{171}$$

$$\left\lfloor A(r+j) + \frac{1}{4} \right\rfloor = f \quad (0 \leq j \leq p), \qquad A(r+p+1) < f - \frac{1}{4}. \tag{172}$$

This theorem sharpens, but does not erase, Theorem 7.20. It closes a real ratio interval; it does not certify the residual compact walls in (169)–(170), and no closure is inferred from their finiteness.

Proof. It is enough to treat

$$\frac{603}{625} \leq \rho \leq \frac{99}{100},$$

because Theorem 7.20 owns the remaining interval $\rho > 99/100$. Put

$$c = \frac{\arccos \rho}{\pi}, \qquad d = d_\rho = 1 - 2c, \qquad \widehat{H}(t) = A(\mu - t), \qquad W = \widehat{H}(0) = G_K(\mu),$$

and set

$$h = f - \frac{1}{4}, \qquad \xi = r + p, \qquad m = n - p, \qquad t_\xi = \mu - \xi.$$

The first shelf and its first drop determine M by

$$\widehat{H}(M) = h, \qquad M \leq t_\xi < M + 1.$$

If level f exists, let $\widehat{H}(T) = f$ and put $\zeta = 0$. If it is absent, put $T = \mu$ and

$$\zeta = f - \widehat{H}(T) \in (0, 1/4].$$

Thus the two level faces remain separate. Since $t_\xi = m + \alpha$ with $\alpha = \mu - q \in [0,1)$, one has $m \geq M - 1$. Changing variables in the exact shelf integral and using the endpoint chord of the convex function $f - \widehat{H}$ gives

$$\boxed{P := R_p + \frac{d}{2}m \geq -\left(\frac{1}{4} + \zeta\right)(T - M) + \frac{d}{2}(M-1).} \tag{173}$$

This is the same wall-safe chord used in (148), now without replacing c, d by $1/20, 9/10$.

Let $\delta = h - W > 0$. The slope bound $0 \leq \widehat{H}' \leq c$ gives

$$M \geq \frac{\delta}{c}. \tag{174}$$

At the rational endpoint, directed evaluation gives

$$\rho \geq \frac{603}{625} \quad\Longrightarrow\quad \theta^2 < \frac{71}{1000}, \qquad c < \bar{c} := \frac{339}{4000}, \qquad \theta = \arccos\rho. \tag{175}$$

Suppose first that $\delta \geq 1$. The shell-increment elasticity (122) gives

$$\frac{\widehat{H}(t_2) - W}{\widehat{H}(t_1) - W} \geq \frac{t_2/t_1}{\sqrt{2t_2/t_1 - 1}}, \qquad 0 < t_1 < t_2 \leq \mu. \tag{176}$$

If level f is present, applying this at M, T gives $T/M \leq 5/2$. Equations (173) and (174) therefore yield

$$\begin{aligned} P &\geq \left(\frac{d}{2} - \frac{3}{8}\right)M - \frac{d}{2} \\ &\geq \frac{1/8 - c}{c} - \left(\frac{1}{2} - c\right) = \frac{1}{8c} - \frac{3}{2} + c. \end{aligned}$$

The final expression decreases on the present range. Hence (175) gives the exact positive ledger

$$\boxed{P > \frac{80921}{1356000}.} \tag{177}$$

If level f is absent, put $a = 1/4 - \zeta \in [0, 1/4]$. Equation (176) gives

$$\frac{\mu/M}{\sqrt{2\mu/M - 1}} \leq 1 + \frac{a}{\delta} \leq 1 + a.$$

With

$$r_*(R) = R^2 + R\sqrt{R^2 - 1},$$

the degree-32 Bernstein certificate already used in (153) proves

$$\left(\frac{1}{2} - a\right)(r_*(1+a) - 1) < \frac{19}{50}, \qquad 0 \leq a \leq \frac{1}{4}.$$

Its exact smallest Bernstein coefficient is $309/396800$. Consequently

$$\begin{aligned} P &> \left(\frac{d}{2} - \frac{19}{50}\right)M - \frac{d}{2} \\ &\geq \frac{3/25 - c}{c} - \left(\frac{1}{2} - c\right) = \frac{3}{25c} - \frac{3}{2} + c. \end{aligned}$$

This expression also decreases on the relevant interval, so

$$\boxed{P > \frac{307}{452000} > 0.} \tag{178}$$

This smaller margin is the limiting ledger in the present argument.

It remains to treat $0 < \delta < 1$. Strict bracketing fixes the initial terminal count without a wall limit:

$$B_0 = [W + \tfrac{1}{4}]_< = f - 1, \qquad W = f - \frac{1}{4} - \delta > f - \frac{5}{4} \geq \frac{3}{4}. \tag{179}$$

Set

$$y = f - W = \delta + \frac{1}{4} \in \left(\frac{1}{4}, \frac{5}{4}\right).$$

We first prove the refined level-distance estimate

$$\boxed{T < \frac{3y}{c}.} \tag{180}$$

It will also show that level f is present.

For fixed ρ, write

$$E_K(t) = \widehat{H}(t) - W = KE_1(t/K).$$

Since E_1 is concave and $E_1(0) = 0$,

$$\partial_K E_K(t) = E_1(u) - uE_1'(u) \geq 0, \qquad u = t/K.$$

Thus, at fixed y, the distance to $W + y$ is largest at the smallest admissible W. Since $W = f - y \geq 2 - y$, it suffices to take

$$f = 2, \qquad W = 2 - y, \qquad x = \frac{y}{W} \in \left[\frac{1}{7}, \frac{5}{3}\right].$$

Put

$$g_\rho = G_1(\rho), \qquad e_\rho(s) = G_1(\rho - s) - \rho G_1(1 - s/\rho) - g_\rho.$$

The concavity needed below is exact, rather than numerical. Indeed, $G_1''(z) = 1/[\pi\sqrt{1 - z^2}]$, and with $b = 1 - s/\rho$,

$$\begin{aligned} e_\rho''(s) &= G_1''(\rho - s) - \frac{1}{\rho} G_1''(1 - s/\rho) \\ &= \frac{1}{\pi\sqrt{1 - \rho^2 b^2}} - \frac{1}{\pi\rho\sqrt{1 - b^2}} \leq 0, \end{aligned}$$

because $\rho^2(1 - b^2) \leq 1 - \rho^2 b^2$. Since $K = W/g_\rho$,

$$\mathcal{F}_\rho(x) := \frac{E_K(3y/c)}{W} - x = \frac{e_\rho(3xg_\rho/c)}{g_\rho} - x \tag{181}$$

is concave in x. It is therefore enough to prove $\mathcal{F}_\rho(x) > 0$ at $x = 1/7, 5/3$.

For the endpoint certificate, let

$$\epsilon = 1 - \rho, \qquad \Phi = \sin\theta - \theta\cos\theta, \qquad W = K\frac{\Phi}{\pi}.$$

For $t_0 = 3y/c$, put $u_0 = t_0/\mu$. Then

$$u_0 = \frac{3x\Phi}{\theta\rho}. \tag{182}$$

The bounds $\Phi < \theta^3/3$ and (175) give

$$u_0 < \frac{5}{3}\frac{71/1000}{603/625} < \frac{1}{8}. \tag{183}$$

There is also the strict lower estimate

$$\Phi > \frac{\rho\theta^3}{3}. \tag{184}$$

Indeed, the difference between the two sides vanishes at zero and has derivative

$$\theta\Phi + \frac{\theta^3 \sin\theta}{3} > 0.$$

Since $\epsilon < \theta^2/2$, (182) and (184) imply

$$\frac{u_0}{\epsilon} > 2x. \tag{185}$$

We now use the noncancelling positive series

$$\arccos(1-z) = \sqrt{2z}\, S(z), \qquad S(z) = \sum_{j\geq 0} \frac{\binom{2j}{j}}{8^j(2j+1)} z^j. \tag{186}$$

If a_j denotes its coefficient, then

$$a_0 = 1, \qquad a_1 = \frac{1}{12}, \qquad a_2 = \frac{3}{160}, \qquad \frac{a_{j+1}}{a_j} = \frac{(2j+1)^2}{4(j+1)(2j+3)} < 1.$$

Thus, for $0 \leq z \leq 1/8$,

$$1 + \frac{z}{12} + \frac{3z^2}{160} \leq S(z) \leq 1 + \frac{z}{12} + \frac{3z^2}{140}. \tag{187}$$

The upper coefficient is exact in direction: monotonicity of a_j gives

$$\sum_{j\geq 2} a_j z^j \leq \frac{3z^2}{160} \sum_{k\geq 0} z^k \leq \frac{3z^2}{160}\frac{8}{7} = \frac{3z^2}{140}.$$

The ball-action identity

$$E_K(t_0) = \frac{\mu}{\pi}\int_0^{u_0} \left[\arccos(\rho(1-u)) - \arccos(1-u)\right] \mathrm{d}u$$

and the substitution $u = \epsilon v$ now give the exact normalized formula

$$\frac{E_K(t_0)}{W} = C_\rho \int_0^{u_0/\epsilon} \left[\sqrt{1+\rho v}\, S(\epsilon(1+\rho v)) - \sqrt{v}\, S(\epsilon v)\right] \mathrm{d}v, \quad C_\rho = \frac{\rho\sqrt{2}\,\epsilon^{3/2}}{\Phi}. \tag{188}$$

Alternating Taylor bounds give

$$\epsilon > \theta^2\left(\frac{1}{2} - \frac{\theta^2}{24}\right), \qquad \Phi < \theta^3\left(\frac{1}{3} - \frac{\theta^2}{30} + \frac{\theta^4}{840}\right).$$

The function

$$t \longmapsto \frac{(1/2 - t/24)^{3/2}}{1/3 - t/30 + t^2/840}$$

is decreasing for $0 \le t \le 71/1000$: after logarithmic differentiation, its sign reduces to

$$84 + 10t - \frac{t^2}{2} > 0.$$

Directed evaluation at $t = 71/1000$ and $\rho = 603/625$ therefore proves

$$\boxed{C_\rho > \frac{36}{25}.} \tag{189}$$

The integrand in (188) is positive. By (185) we may retain only $0 \le v \le V := 2x$. On this interval,

$$\epsilon v \le \frac{22}{625}\frac{10}{3} = \frac{44}{375} < \frac{1}{8}.$$

Keeping the first three positive terms in the first copy of S and using the upper bound in (187) in the subtracted copy produces

$$\begin{aligned}\mathcal{I}_\rho(V) = {} & \frac{2}{3\rho}\Big((1+\rho V)^{3/2} - 1\Big) \\ & + \frac{\epsilon}{30\rho}\Big((1+\rho V)^{5/2} - 1\Big) \\ & + \frac{3\epsilon^2}{560\rho}\Big((1+\rho V)^{7/2} - 1\Big) \\ & - \frac{2}{3}V^{3/2} - \frac{\epsilon}{30}V^{5/2} - \frac{3\epsilon^2}{490}V^{7/2}.\end{aligned} \tag{190}$$

Partitioning $[603/625, 99/100]$ into 16 equal rational panels, a directed Arb evaluation on each entire panel, at each of $x = 1/7, 5/3$, proves

$$\mathcal{I}_\rho(2x) > 0, \qquad \frac{36}{25}\mathcal{I}_\rho(2x) - x > 0. \tag{191}$$

These are interval enclosures of all ratios in each panel, not point samples. At 384, 768, and 1024 bits, the smallest directed lower margin in the second inequality is $0.122748159345\ldots$, and the smallest retained-integral lower bound is $0.202999800557\ldots$. Equations (188)–(191), followed by concavity in (181), prove (180).

Level f is therefore present, so $\zeta = 0$. From (173), (174), and (180),

$$\begin{aligned} cP &> -\frac{3}{4}\left(\delta + \frac{1}{4}\right) + \left(\frac{1}{4} + \frac{d}{2}\right)\delta - \frac{cd}{2} \\ &= -c\delta - \frac{3}{16} - \frac{cd}{2} > -c - \frac{3}{16} - \frac{cd}{2}.\end{aligned} \tag{192}$$

At the terminal start, strict bracketing gives

$$B = [G_K(q) + \tfrac{1}{4}]_< \ge f - 1, \qquad Q = [A(q) + \tfrac{1}{4}]_< \le f - 1.$$

For every $1 \le k \le f - 1$,

$$k - \frac{1}{4} \le f - \frac{5}{4} < W = G_K(\mu)$$

strictly. Hence every retained ball angle in Corollary 7.4 is smaller than $\theta = \pi c$. That reserve and the cap bound (30) give

$$cD_A(q) > \frac{1}{2} - \frac{7}{5}c. \tag{193}$$

Adding this to (192), and using $d = 1 - 2c$, yields

$$c\mathcal{S} > \frac{5}{16} - \frac{29}{10}c + c^2.$$

The right side decreases on the present range. At $c = \bar{c}$,

$$\boxed{c\mathcal{S} > \frac{1182521}{16000000} > 0.} \tag{194}$$

Together with (177) and (178), this proves (168).

It remains to derive the residual box. For $\rho < 603/625$, monotonicity of G_1 and directed evaluation at the rational endpoint give

$$\eta_\rho > \frac{1}{504}, \qquad \eta_\rho < \frac{1}{9}. \tag{195}$$

The second inequality also follows from the previously used bound $G_1(1/2) < 1/9$. Moreover, using $\pi < 355/113$,

$$a_\rho < 2\frac{355}{113}\frac{603}{22} = \frac{214065}{1243}.$$

Since $C_* < 13/10$, (195) implies

$$a_\rho + 2\eta_\rho C_* < \frac{214065}{1243} + \frac{13}{45} < 173. \tag{196}$$

Because $C_* > 1/2$,

$$\sqrt{a_\rho^2 + 4a_\rho\eta_\rho C_* + \eta_\rho^2} < a_\rho + 2\eta_\rho C_*.$$

The fixed-ratio threshold (42) therefore obeys

$$\boxed{K_0(\rho) < 173 \cdot 504^2 = 43944768.} \tag{197}$$

Negativity excludes $p = 0$, so $p \geq 1$ and $n \geq 2$. Since $r < \mu < K < 43944768$ and $r \geq 1/2$,

$$\rho > \frac{1}{87889536}, \qquad 2r \leq 87889535, \qquad n \leq 43944767.$$

As before, $A(r) \geq 7/4$ and $A(0) > A(r)$ give $K - \mu > 7\pi/4$ and $K > 21/4$. Finally, $\pi > 3$ gives

$$f, B < \frac{K}{3} + \frac{1}{4} < 14648256 + \frac{1}{4},$$

so the integer bounds in (170) follow. The shelf drop gives $Q \leq f - 1$, while $A \leq G_K$ gives $Q \leq B$; the remaining chamber conditions are their ordinary-floor and strict-bracket definitions.

All finite comparisons just used are replayed by the Round 8A verifier

`scripts/general_d_high_floor_fixed_ratio_verify.py`.

Independent directed runs at 384, 768, and 1024 bits give the same exact ledgers and cover the full rational panels. This proves (169)–(172). □

Remark 7.22 (Limiting ledger in the extended ratio argument). The present-level bound remains positive until

$$c < \frac{3-\sqrt{7}}{4} = 0.08856\ldots,$$

and (194) remains positive farther still. The first obstruction is the absent-level estimate: its lower bound changes sign at

$$c_{\text{abs}} = \frac{\frac{3}{2} - \sqrt{177/100}}{2} = 0.0847932652\ldots, \qquad \rho_{\text{abs}} = \cos(\pi c_{\text{abs}}) = 0.9647285943\ldots.$$

The endpoint $603/625 = 0.9648$ lies only $7.14 \cdot 10^{-5}$ above this barrier. Pushing lower requires an improvement of the absent-level chord loss $19/50$ or additional positive terminal information on that face. This is an obstruction to the present proof ledger, not a counterexample to the shell scalar.

Theorem 7.23 (Terminally strengthened high-floor exclusion)**.** *In the high-floor first-drop branch, suppose*

$$f = F_0 \geq 2, \qquad p < n, \qquad \frac{477}{500} \leq \rho < 1.$$

Then the reduced scalar is strictly positive:

$$\boxed{\mathcal{S} = D_A(q) + R_p + \frac{d_\rho}{2}(n-p) > 0.} \tag{198}$$

Consequently every remaining negative high-floor first-drop candidate satisfies

$$\boxed{\frac{1}{29931928} < \rho < \frac{477}{500}, \qquad \frac{21}{4} < K < 14965964,} \tag{199}$$

and the exact finite integer ranges

$$\boxed{\begin{array}{c} r \in \tfrac{1}{2}\mathbb{N}, \qquad 1 \leq 2r \leq 29931927, \\ 2 \leq n \leq 14965963, \qquad 1 \leq p \leq n-1, \\ 2 \leq f \leq 4988654, \\ 0 \leq Q \leq f-1, \qquad Q \leq B \leq 4988654. \end{array}} \tag{200}$$

The chamber conditions

$$\mu = \rho K, \qquad q = r + n \leq \mu < q+1, \qquad K - \mu > \frac{7\pi}{4}, \tag{201}$$

$$\left\lfloor A(r+j) + \frac{1}{4} \right\rfloor = f \quad (0 \leq j \leq p), \qquad A(r+p+1) < f - \frac{1}{4} \tag{202}$$

remain in force. This theorem supersedes the residual box in Theorem 7.21; it does not certify the compact walls left by (199).

Proof. It is enough to treat

$$\frac{477}{500} \leq \rho \leq \frac{99}{100},$$

because Theorem 7.20 owns $\rho > 99/100$. Retain the notation

$$c = \frac{\arccos \rho}{\pi}, \qquad d = d_\rho = 1 - 2c, \qquad \widehat{H}(t) = A(\mu - t), \qquad W = \widehat{H}(0) = G_K(\mu),$$

$$h = f - \frac{1}{4}, \qquad \xi = r + p, \qquad m = n - p, \qquad \delta = h - W > 0.$$

The endpoint-chord construction in (173) and (174) remains unchanged:

$$P := R_p + \frac{d}{2}m \geq -\left(\frac{1}{4} + \zeta\right)(T - M) + \frac{d}{2}(M-1), \qquad M \geq \frac{\delta}{c}. \tag{203}$$

Here $\widehat{H}(M) = h$; if level f exists then $\widehat{H}(T) = f$ and $\zeta = 0$, while on the absent face $T = \mu$ and $\zeta = f - \widehat{H}(T) \in (0, 1/4]$. Directed endpoint evaluation gives

$$\rho \geq \frac{477}{500} \implies (\arccos \rho)^2 < \frac{93}{1000}, \qquad c < \bar{c} := \frac{97}{1000}. \tag{204}$$

Suppose first that $\delta \geq 1$ and put

$$\gamma = \frac{3}{25} - c > 0.$$

On the present-level face, the elasticity estimate (176) gives $T/M \leq 5/2$. Thus (203) yields

$$cP \geq \left(\frac{1}{8} - c\right)\delta - \frac{cd}{2} > \gamma\delta - \frac{cd}{2},$$

where $1/8 - 3/25 = 1/200$. On the absent-level face, the degree-32 Bernstein comparison with smallest coefficient $309/396800$ gives

$$\left(\frac{1}{2} - a\right)(r_*(1+a) - 1) < \frac{19}{50}, \qquad 0 \leq a \leq \frac{1}{4}.$$

The same calculation as in (178), now retaining δ, gives the common strict prefix

$$\boxed{cP > \gamma\delta - \frac{cd}{2}.} \tag{205}$$

At the terminal start define

$$B_0 = [W + \tfrac{1}{4}]_<, \qquad B = [G_K(q) + \tfrac{1}{4}]_<, \qquad Q = [A(q) + \tfrac{1}{4}]_<.$$

Since $q = \mu - \alpha$ with $0 \leq \alpha < 1$, the slope bound gives $0 \leq A(q) - W < c$. Hence, on all strict walls,

$$B \geq B_0, \qquad Q \leq B_0 + 1. \tag{206}$$

If $B_0 \geq 1$, the first B_0 ball angles are all smaller than πc. The exact-angle reserve and the cap bound (30) give

$$cD_A(q) > B_0\left(\frac{1}{2} - c\right) - \frac{7c}{5}.$$

Combining this with (205), $\delta \geq 1$, and $B_0 \geq 1$ yields

$$c\mathcal{S} > \frac{31}{50} - \frac{39}{10}c + c^2 \geq \frac{251109}{1000000} > 0. \tag{207}$$

It remains in the $\delta \geq 1$ sector to consider $B_0 = 0$. Strict bracketing says

$$0 < W \leq \frac{3}{4}, \qquad Q \in \{0, 1\}.$$

Moreover $q+1>\mu$, so every sample after q lies on the outer ball tail and has strict count zero. If $Q=0$, the shell discrete tail is empty. The outer tangent triangle gives

$$D_A(q) \geq 2\int_{\mu}^{K} G_K(z)\,dz \geq \frac{W^2}{c}.$$

Since $\delta \geq 7/4 - W$, minimizing the resulting convex quadratic at $W=\gamma/2$ gives

$$c\mathcal{S} > \frac{129}{625} - \frac{219}{100}c + \frac{3}{4}c^2 \geq \frac{4107}{4000000} > 0. \tag{208}$$

This is the limiting ledger of the theorem.

If $Q=1$, then $A(q)>3/4$. Since $A(q)-W<c\alpha$,

$$\alpha > \frac{3/4 - W}{c}, \qquad \frac{3}{4} - c < W \leq \frac{3}{4}.$$

The inner payment $A \geq W$ on $[q,\mu]$, together with the outer tangent triangle, gives

$$cD_A(q) > \frac{3}{2}W - W^2 - c.$$

After (205) and $\delta \geq 7/4 - W$ are added, the expression is concave in W. Its two endpoint lower bounds are

$$\frac{273}{400} - \frac{5}{2}c + c^2 \geq \frac{449409}{1000000}, \qquad \frac{273}{400} - \frac{119}{50}c - c^2 \geq \frac{442231}{1000000}. \tag{209}$$

Thus every $\delta \geq 1$ face is positive.

Now suppose $0<\delta<1$. Then

$$B_0 = f-1, \qquad W = f - \frac{1}{4} - \delta > f - \frac{5}{4} \geq \frac{3}{4}, \qquad y = f - W = \delta + \frac{1}{4}.$$

We need the enlarged level-distance estimate

$$\boxed{T < \frac{3y}{c}.} \tag{210}$$

The scale reduction and concavity argument in (181) is unchanged: it reduces the claim to $f=2$, $W=2-y$, and the two endpoint values

$$x = \frac{y}{W} \in \left[\frac{1}{7}, \frac{5}{3}\right].$$

For $t_0 = 3y/c$ and $u_0 = t_0/\mu$, equations (182) and (184) now give

$$u_0 < \frac{5}{3}\frac{93/1000}{477/500} < \frac{1}{6}, \qquad \frac{u_0}{1-\rho} > 2x.$$

Write $\epsilon = 1-\rho$ and use the positive series (186). On the enlarged subtracted range

$$0 \leq z \leq \frac{23}{150}$$

its decreasing coefficients give

$$1 + \frac{z}{12} + \frac{3z^2}{160} \leq S(z) \leq 1 + \frac{z}{12} + \frac{45z^2}{2032}. \tag{211}$$

The positive first copy of the series is used only for $z \le 299/1500 < 1$. The coefficient in the noncancelling identity (188) satisfies

$$C_\rho = \frac{\rho\sqrt{2}\,\epsilon^{3/2}}{\sin\theta - \theta\cos\theta} > \frac{7}{5}. \tag{212}$$

Indeed, the same decreasing Taylor quotient as in (189), evaluated at $\theta^2 = 93/1000$ and $\rho = 477/500$, has this directed lower bound.

Retaining $0 \le v \le V = 2x$ in the positive integral produces

$$\begin{aligned}\mathcal{I}^{\mathrm{B}}_\rho(V) = {}& \frac{2}{3\rho}\left((1+\rho V)^{3/2} - 1\right) + \frac{\epsilon}{30\rho}\left((1+\rho V)^{5/2} - 1\right) \\ & + \frac{3\epsilon^2}{560\rho}\left((1+\rho V)^{7/2} - 1\right) - \frac{2}{3}V^{3/2} - \frac{\epsilon}{30}V^{5/2} - \frac{45\epsilon^2}{7112}V^{7/2}.\end{aligned} \tag{213}$$

The last coefficient is exactly $(45/2032)(2/7)$. Partitioning $[477/500, 99/100]$ into 32 equal rational panels and evaluating each entire Arb interval at $x = 1/7, 5/3$ gives

$$\mathcal{I}^{\mathrm{B}}_\rho(2x) > 0, \qquad \frac{7}{5}\mathcal{I}^{\mathrm{B}}_\rho(2x) - x > 0. \tag{214}$$

The smallest directed lower bounds are 0.2031479283 for the retained integral and 0.0607609473 for the second expression. Concavity in x proves (210).

Level f is therefore present. The same chord calculation as in (192) gives

$$cP > -c - \frac{3}{16} - \frac{cd}{2}.$$

The shelf drop (202) gives $Q \le f-1$, while $B_0 = f-1$ supplies $f-1$ angles smaller than πc. Hence

$$cD_A(q) > \frac{1}{2} - \frac{7}{5}c.$$

Adding and using $d = 1-2c$ yields

$$c\mathcal{S} > \frac{5}{16} - \frac{29}{10}c + c^2 \ge \frac{40609}{1000000} > 0. \tag{215}$$

Together with (207)–(209), this proves (198).

For the new box, a residual ratio $\rho < 477/500$ satisfies

$$\eta_\rho > \frac{1}{338}, \qquad \eta_\rho < \frac{1}{9}, \qquad a_\rho < 2\,\frac{355}{113}\,\frac{477}{23} = \frac{338670}{2599}.$$

Since $C_* < 13/10$,

$$a_\rho + 2\eta_\rho C_* < 131.$$

The square root in (42) is smaller than this same quantity because $C_* > 1/2$. Therefore

$$\boxed{K_0(\rho) < 131 \cdot 338^2 = 14965964.} \tag{216}$$

The arguments following (197) now give $2r \le 29931927$, $n \le 14965963$, and $f, B \le 4988654$, together with (201)–(202).

All directed and exact comparisons in this proof are replayed by `scripts/general_d_high_floor_absent_terminal_verify.py`.

Independent runs at 384, 512, 768, and 1024 bits give the same rational ledgers and cover all 64 ratio-panel endpoint evaluations. $\square$

Remark 7.24 (The new limiting zero-count ledger). The weakest polynomial in the strengthened argument is (208). Its first zero is

$$c_* = \frac{219 - \sqrt{41769}}{150} = 0.0975022935\ldots, \qquad \cos(\pi c_*) = 0.9534519995\ldots.$$

The rational choice $477/500 = 0.954$ remains strictly above this point. The high-floor active-width relation contains additional information not used in the zero-count estimate, so this sign change is an obstruction to the present ledger, not a counterexample to the scalar.

Theorem 7.25 (Active-width and sharp-cap high-floor exclusion)**.** *In the high-floor first-drop branch, suppose*

$$f = F_0 \geq 2, \qquad p < n, \qquad \frac{93}{100} \leq \rho < 1.$$

Then

$$\boxed{\mathcal{S} = D_A(q) + R_p + \frac{d_\rho}{2}(n-p) > 0.} \tag{217}$$

Consequently every remaining negative high-floor first-drop candidate satisfies

$$\boxed{\frac{1}{5443200} < \rho < \frac{93}{100}, \qquad \frac{21}{4} < K < 2721600,} \tag{218}$$

and

$$\boxed{\begin{array}{c} r \in \frac{1}{2}\mathbb{N}, \qquad 1 \leq 2r \leq 5443199, \\ 2 \leq n \leq 2721599, \qquad 1 \leq p \leq n-1, \\ 2 \leq f \leq 907200, \\ 0 \leq Q \leq f-1, \qquad Q \leq B \leq 907200. \end{array}} \tag{219}$$

The exact chamber conditions remain

$$\mu = \rho K, \qquad q = r + n \leq \mu < q + 1, \qquad K - \mu > \frac{7\pi}{4}, \tag{220}$$

$$\left\lfloor A(r+j) + \frac{1}{4} \right\rfloor = f \quad (0 \leq j \leq p), \qquad A(r+p+1) < f - \frac{1}{4}. \tag{221}$$

This theorem supersedes the residual box in Theorem 7.23. It certifies the displayed ratio interval, not the compact walls left by (218)*.*

Proof. By Theorem 7.23, it is enough to treat

$$\frac{93}{100} \leq \rho \leq \frac{477}{500}.$$

Retain the notation

$$\epsilon = 1 - \rho, \qquad a = \epsilon K, \qquad h = f - \frac{1}{4}, \qquad W = G_K(\mu), \qquad \delta = h - W,$$

$$c = \frac{\arccos \rho}{\pi}, \qquad d = 1 - 2c, \qquad g_\rho = G_1(\rho).$$

The active high-floor inequality (107) and $W = ag_\rho/\epsilon$ give, with

$$\lambda_\rho = \frac{\pi g_\rho}{\epsilon},$$

$$W > \lambda_\rho(W + \delta). \tag{222}$$

Thus $\lambda_\rho < 1$. Moreover

$$\pi g_\rho = \int_0^\epsilon \arccos(1 - y)\, \mathrm{d}y > \frac{2\sqrt{2}}{3}\epsilon^{3/2}.$$

Since $\epsilon \geq 23/500$ and

$$\frac{8}{9}\frac{23}{500} > \frac{1}{25},$$

one has $\lambda_\rho > 1/5$. As $\delta \geq 7/4 - W$, (222) gives

$$\boxed{W > \frac{7}{4}\lambda_\rho > \frac{7}{20}.} \tag{223}$$

Directed endpoint evaluation also gives

$$(\arccos \rho)^2 < \frac{71}{500}, \qquad c < \bar{c} := \frac{1199}{10000} < \frac{3}{25}. \tag{224}$$

Suppose first that $\delta \geq 1$. The endpoint-chord and degree-32 absent-level arguments used in (205) remain valid and give

$$cP > \left(\frac{3}{25} - c\right)\delta - \frac{cd}{2}, \qquad P = R_p + \frac{d}{2}(n - p). \tag{225}$$

The exact smallest Bernstein coefficient is still 309/396800. For $B_0 \geq 1$, the Round 8B terminal reserve gives

$$c\mathcal{S} > \frac{31}{50} - \frac{39}{10}c + c^2 \geq \frac{16676601}{100000000} > 0.$$

If $B_0 = Q = 0$, the outer tangent triangle, together with (223), gives

$$\begin{aligned} c\mathcal{S} &> \left(\frac{3}{25} - c\right)\left(\frac{7}{4} - W\right) - \frac{c}{2} + c^2 + W^2 \\ &\geq \frac{7706601}{100000000} > 0. \end{aligned}$$

Indeed, the derivative in $W \geq 7/20$ is larger than $7/10 - 3/25 = 29/50$. Finally, on the face $B_0 = 0, Q = 1$, the two endpoint ledgers are

$$\frac{273}{400} - \frac{5}{2}c + c^2 \geq \frac{39712601}{100000000} > 0,$$

$$\frac{273}{400} - \frac{119}{50}c - c^2 \geq \frac{38276199}{100000000} > 0.$$

This proves positivity on every $\delta \geq 1$ face, including the strict walls $\delta = 1$, $W = 3/4$, and $A(q) = 3/4$.

Now let $0 < \delta < 1$. As in Round 8B,

$$B_0 = f - 1, \qquad W = f - \frac{1}{4} - \delta > f - \frac{5}{4} \geq \frac{3}{4}, \qquad y = f - W = \delta + \frac{1}{4}.$$

We first prove

$$\boxed{T < \frac{3y}{c}.} \tag{226}$$

The scale and concavity reductions in (181) reduce the claim to $f = 2$ and the endpoints

$$x = \frac{y}{W} \in \left\{\frac{1}{7}, \frac{5}{3}\right\}.$$

For $t_0 = 3y/c$ and $u_0 = t_0/\mu$, the estimates used in (185) give

$$u_0 < \frac{5}{3}\frac{71/500}{93/100} = \frac{71}{279} < \frac{51}{200} < 1, \qquad \frac{u_0}{1-\rho} > 2x.$$

Use the positive series

$$\arccos(1-z) = \sqrt{2z}\, S(z), \qquad S(z) = \sum_{j\geq 0} \frac{\binom{2j}{j}}{8^j(2j+1)} z^j.$$

On the subtracted retained range $z \leq 7/30$, its decreasing coefficients give

$$1 + \frac{z}{12} + \frac{3z^2}{160} \leq S(z) \leq 1 + \frac{z}{12} + \frac{9z^2}{368}, \tag{227}$$

where $(3/160)/(1-7/30) = 9/368$. The positive first copy is used only for $z \leq 91/300 < 1$. The decreasing Taylor quotient from (212), evaluated at $\rho = 93/100$ and $\theta^2 = 71/500$, gives

$$C_\rho = \frac{\rho\sqrt{2}(1-\rho)^{3/2}}{\sin\theta - \theta\cos\theta} > \frac{1389}{1000}. \tag{228}$$

Retaining $0 \leq v \leq V = 2x$ therefore produces

$$\begin{aligned} \mathcal{I}_\rho^{\mathrm{D}}(V) = {} & \frac{2}{3\rho}\Big((1+\rho V)^{3/2} - 1\Big) + \frac{1-\rho}{30\rho}\Big((1+\rho V)^{5/2} - 1\Big) \\ & + \frac{3(1-\rho)^2}{560\rho}\Big((1+\rho V)^{7/2} - 1\Big) - \frac{2}{3}V^{3/2} - \frac{1-\rho}{30}V^{5/2} - \frac{9(1-\rho)^2}{1288}V^{7/2}. \end{aligned} \tag{229}$$

The last coefficient is $(9/368)(2/7)$. On 64 equal rational panels covering $[93/100, 477/500]$, directed evaluation at $x = 1/7, 5/3$ gives

$$\mathcal{I}_\rho^{\mathrm{D}}(2x) > 0, \qquad \frac{1389}{1000}\mathcal{I}_\rho^{\mathrm{D}}(2x) - x > 0. \tag{230}$$

Across runs at 384, 512, 768, and 1024 bits, the smallest retained-integral lower bound exceeds 0.2037148927. The smallest directed margin exceeds 0.0162868443. Concavity proves (226).

It remains to sharpen the terminal cap. Since $p < n$, one has $n \geq 1$ and $q = r + n \geq 3/2$. If $q = \mu$, the cap vanishes. If $q < \mu$, strict convexity gives

$$2\int_q^\mu G_\mu(z)\,\mathrm{d}z < G_\mu(q)(\mu - q) < G_{q+1}(q).$$

For $H(q) = G_{q+1}(q)$ and $q/(q+1) = \cos\vartheta$,

$$H'(q) = \frac{\sin\vartheta - \vartheta}{\pi} < 0.$$

Directed evaluation at the endpoint gives

$$\boxed{2\int_q^\mu G_\mu(z)\,\mathrm{d}z < G_{5/2}(3/2) < \frac{1}{5}.} \tag{231}$$

The endpoint chord and (226) yield

$$cP > -c - \frac{3}{16} - \frac{cd}{2}.$$

The exact-angle reserve, $B_0 = f - 1$, $Q \le f - 1$, and (231) give

$$cD_A(q) > \frac{f-1}{2} - c(f-1) - \frac{c}{5} \ge \frac{1}{2} - \frac{6c}{5}.$$

Adding and using $d = 1 - 2c$ proves

$$c\mathcal{S} > \frac{5}{16} - \frac{27}{10}c + c^2 \ge \frac{314601}{100000000} > 0. \tag{232}$$

Together with the $\delta \ge 1$ ledgers, this proves (217).

For a residual ratio $\rho < 93/100$, monotonicity and directed endpoint evaluation give

$$\eta_\rho > \frac{1}{180}, \qquad \eta_\rho < \frac{1}{9},$$

and

$$a_\rho < 2\frac{355}{113}\frac{93}{7} = \frac{66030}{791}.$$

Since $C_* < 13/10$,

$$a_\rho + 2\eta_\rho C_* < \frac{66030}{791} + \frac{13}{45} = \frac{2981633}{35595} < 84.$$

The fixed-ratio threshold therefore satisfies

$$\boxed{K_0(\rho) < 84 \cdot 180^2 = 2721600.} \tag{233}$$

The integer arguments following (216) give (218)–(219) and the chamber conditions.

All directed and exact comparisons in this proof are replayed by

`scripts/general_d_high_floor_sharp_cap_verify.py`.

The helper hash, the full 128 panel-endpoint evaluations, the cap endpoint, every rational ledger, and the cutoff arithmetic are checked inside the certificate. □

Theorem 7.26 (Refined-cap high-floor exclusion)**.** *In the high-floor first-drop branch, suppose*

$$f = F_0 \ge 2, \qquad p < n, \qquad \frac{927}{1000} \le \rho < 1.$$

Then

$$\boxed{\mathcal{S} = D_A(q) + R_p + \frac{d_\rho}{2}(n-p) > 0.} \tag{234}$$

Consequently every remaining negative high-floor first-drop candidate satisfies

$$\boxed{\frac{1}{4626882} < \rho < \frac{927}{1000}, \qquad \frac{21}{4} < K < 2313441,} \tag{235}$$

and

$$\boxed{\begin{gathered} r \in \tfrac{1}{2}\mathbb{N}, \qquad 1 \le 2r \le 4626881, \\ 2 \le n \le 2313440, \qquad 1 \le p \le n-1, \\ 2 \le f \le 771147, \\ 0 \le Q \le f-1, \qquad Q \le B \le 771147. \end{gathered}} \tag{236}$$

The exact chamber conditions remain

$$\mu = \rho K, \qquad q = r + n \le \mu < q+1, \qquad K - \mu > \frac{7\pi}{4}, \tag{237}$$

$$\left\lfloor A(r+j) + \frac{1}{4} \right\rfloor = f \quad (0 \le j \le p), \qquad A(r+p+1) < f - \frac{1}{4}. \tag{238}$$

This theorem supersedes the residual box in Theorem 7.25. It certifies the displayed ratio interval, not the compact walls left by (235).

Proof. By Theorem 7.25, it is enough to treat

$$\frac{927}{1000} \le \rho \le \frac{93}{100}.$$

If $p = 0$, then $R_0 = 0$, and Theorem 6.7 gives $\mathcal{S} = D_A(q) + d_\rho n/2 > 0$. A negative candidate must therefore have $p \ge 1$. Since $p < n$ and $r \ge 1/2$, this implies

$$n \ge 2, \qquad q = r + n \ge \frac{5}{2}. \tag{239}$$

If $q = \mu$, the inner cap vanishes. Otherwise strict convexity of G_μ gives

$$2\int_q^\mu G_\mu(z)\,\mathrm{d}z < G_\mu(q)(\mu - q) < G_{q+1}(q).$$

For $H(q) = G_{q+1}(q)$ and $q/(q+1) = \cos\vartheta$,

$$H'(q) = \frac{\sin\vartheta - \vartheta}{\pi} < 0.$$

Directed evaluation at the endpoint in (239) therefore proves

$$\boxed{2\int_q^\mu G_\mu(z)\,\mathrm{d}z < G_{7/2}(5/2) < \frac{1}{6}.} \tag{240}$$

Retain

$$c = \frac{\arccos\rho}{\pi}, \qquad d = 1 - 2c, \qquad W = G_K(\mu), \qquad h = f - \frac{1}{4}, \qquad \delta = h - W.$$

We first sharpen the absent-level endpoint chord used in Round 8D. When level f is absent, put $a = 1/4 - \zeta \in [0, 1/4]$ and

$$r_*(R) = R^2 + R\sqrt{R^2 - 1}.$$

The elasticity reduction for $\delta \geq 1$ contains the coefficient

$$L(a) = \left(\frac{1}{2} - a\right)\left(r_*(1+a) - 1\right).$$

Set

$$b(a) = \frac{377}{1000} - \left(\frac{1}{2} - a\right)(2a + a^2) = \frac{377}{1000} - a + \frac{3}{2}a^2 + a^3.$$

The function b decreases on $[0, 1/4]$ and $b(1/4) = 1891/8000 > 0$. Moreover

$$\begin{aligned} &b(a)^2 - \left(\frac{1}{2} - a\right)^2 (1+a)^2 a(2+a) \\ &= \frac{142129}{10^6} - \frac{627}{500}a + \frac{2881}{1000}a^2 - \frac{123}{500}a^3 - a^4 > 0. \end{aligned} \tag{241}$$

Indeed, after $a = y/4$, all degree-48 Bernstein coefficients are positive. Their smallest member is

$$\frac{7422203}{77832000000}.$$

The sign of b makes the squaring step legitimate, so

$$\boxed{L(a) < \frac{377}{1000}.} \tag{242}$$

Put

$$\beta = \frac{123}{1000} - c.$$

If level f is present, the Round 8B elasticity estimate gives

$$cP \geq \left(\frac{1}{8} - c\right)\delta - \frac{cd}{2} > \beta\delta - \frac{cd}{2}.$$

If it is absent, (242) gives

$$P > \left(\frac{d}{2} - \frac{377}{1000}\right)M - \frac{d}{2} = \beta M - \frac{d}{2}.$$

Directed endpoint evaluation on the present ratio band gives

$$(\arccos\rho)^2 < \frac{37}{250}, \qquad c < \bar{c} := \frac{49}{400} < \frac{123}{1000}. \tag{243}$$

Thus $\beta > 0$, and $M \geq \delta/c$ yields the common prefix

$$\boxed{cP > \beta\delta - \frac{cd}{2}.} \tag{244}$$

The active-width relation still gives

$$W > \lambda_\rho(W + \delta), \qquad \lambda_\rho = \frac{\pi G_1(\rho)}{1-\rho}.$$

Here $\epsilon = 1 - \rho \geq 7/100$, and

$$\lambda_\rho > \frac{2\sqrt{2}}{3}\sqrt{\epsilon} > \frac{249}{1000}, \qquad \frac{8}{9}\frac{7}{100} - \left(\frac{249}{1000}\right)^2 = \frac{1991}{9000000} > 0.$$

Since $W + \delta = f - 1/4 \ge 7/4$,

$$\boxed{W > \frac{1743}{4000}.} \tag{245}$$

Suppose first that $\delta \ge 1$. On the face $B_0 \ge 1$, the exact-angle reserve, $Q \le B_0 + 1$, and (240) give

$$cD_A(q) > B_0\left(\frac{1}{2} - c\right) - \frac{7c}{6}.$$

Combining this with (244), $B_0 \ge 1$, and $\delta \ge 1$, gives

$$c\mathcal{S} > \frac{623}{1000} - \frac{11}{3}c + c^2 \ge \frac{90643}{480000} > 0. \tag{246}$$

If $B_0 = Q = 0$, the shell tail is empty and the outer tangent triangle gives $cD_A(q) \ge W^2$. Since $\delta \ge 7/4 - W$, equations (244) and (245) give

$$c\mathcal{S} > \beta\left(\frac{7}{4} - W\right) - \frac{c}{2} + c^2 + W^2 \ge \frac{2308663}{16000000} > 0. \tag{247}$$

The derivative in W is $2W - \beta > 0$, and the last expression decreases in c, which justifies the displayed endpoints. Finally, if $B_0 = 0, Q = 1$, then $3/4 - c < W \le 3/4$ and

$$cD_A(q) > \frac{3}{2}W - W^2 - c.$$

The resulting expression is concave in W. Its two endpoint ledgers are

$$\frac{1371}{2000} - \frac{5}{2}c + c^2 \ge \frac{63081}{160000} > 0, \qquad \frac{1371}{2000} - \frac{2377}{1000}c - c^2 \ge \frac{303449}{800000} > 0. \tag{248}$$

This closes every $\delta \ge 1$ terminal face, including the strict-count walls.

Now let $0 < \delta < 1$. The scale and concavity reductions used in (226) again reduce

$$T < \frac{3y}{c}, \qquad y = f - W = \delta + \frac{1}{4},$$

to $f = 2$ and $x = y/W \in \{1/7, 5/3\}$. With $t_0 = 3y/c$ and $u_0 = t_0/\mu$, equation (243) gives

$$u_0 < \frac{5}{3}\frac{37/250}{927/1000} = \frac{740}{2781} < \frac{267}{1000} < 1, \qquad \frac{u_0}{1-\rho} > 2x.$$

On the subtracted retained range, $z \le (73/1000)(10/3) = 73/300$, so the positive arccos series satisfies

$$1 + \frac{z}{12} + \frac{3z^2}{160} \le S(z) \le 1 + \frac{z}{12} + \frac{45z^2}{1816}. \tag{249}$$

The positive first copy is used only for $z \le 949/3000 < 1$. The decreasing Taylor quotient from Round 8D, now evaluated at $\rho = 927/1000$ and $\theta^2 = 37/250$, gives

$$C_\rho = \frac{\rho\sqrt{2}(1-\rho)^{3/2}}{\sin\theta - \theta\cos\theta} > \frac{173}{125}.$$

For $V = 2x$, the retained lower integral is

$$\begin{aligned}\mathcal{I}^{\mathrm{E}}_\rho(V) = {}& \frac{2}{3\rho}\left((1+\rho V)^{3/2} - 1\right) + \frac{1-\rho}{30\rho}\left((1+\rho V)^{5/2} - 1\right) \\ &+ \frac{3(1-\rho)^2}{560\rho}\left((1+\rho V)^{7/2} - 1\right) - \frac{2}{3}V^{3/2} - \frac{1-\rho}{30}V^{5/2} \\ &- \frac{45(1-\rho)^2}{6356}V^{7/2}.\end{aligned} \tag{250}$$

On 64 equal rational panels covering $[927/1000, 93/100]$, directed evaluation at both values of x proves

$$\mathcal{I}_\rho^{\mathrm{E}}(2x) > 0, \qquad \frac{173}{125}\mathcal{I}_\rho^{\mathrm{E}}(2x) - x > 0.$$

Across 384, 512, 768, and 1024 bits, the smallest directed lower bounds are 0.2040442323 and 0.0085053826, respectively. Concavity proves $T < 3y/c$ on the whole band. Consequently

$$cP > -c - \frac{3}{16} - \frac{cd}{2}.$$

Here $B_0 = f - 1$ and $Q \le f - 1$, so the refined cap gives

$$cD_A(q) > \frac{f-1}{2} - c(f-1) - \frac{c}{6} \ge \frac{1}{2} - \frac{7c}{6}.$$

Adding these inequalities and using $d = 1 - 2c$ gives

$$c\mathcal{S} > \frac{5}{16} - \frac{8}{3}c + c^2 \ge \frac{403}{480000} > 0. \tag{251}$$

Together with (246)–(248), this proves (234).

For a residual ratio $\rho < 927/1000$, monotonicity and directed endpoint evaluation give

$$\eta_\rho > \frac{1}{169}, \qquad \eta_\rho < \frac{1}{9},$$

and

$$a_\rho < 2\frac{355}{113}\frac{927}{73} = \frac{658170}{8249}.$$

Since $C_* < 13/10$,

$$a_\rho + 2\eta_\rho C_* < \frac{658170}{8249} + \frac{13}{45} = \frac{29724887}{371205} < 81.$$

As $C_* > 1/2$, the square root in the fixed-ratio threshold is smaller than the same left side. Hence

$$\boxed{K_0(\rho) < 81 \cdot 169^2 = 2313441.} \tag{252}$$

The integer arguments following (233) give (235)–(236) and the chamber conditions.

All exact and directed comparisons in this proof are replayed by

`scripts/general_d_high_floor_refined_cap_verify.py`.

The certificate pins the Round 8D verifier hash, checks the degree-48 Bernstein ledger, and covers all 128 panel-endpoint evaluations at four precisions. □

7.3 Global finite exhaustion of the no-drop branch

We first record the root-free consequence of the terminal estimate that drives the exhaustion. For $J \ge 1$, put

$$C_0^3 = \frac{\pi}{12}, \qquad c_* = \frac{4\sqrt{2}}{15}, \qquad S_J = \sum_{k=1}^{J} (k - \tfrac{1}{4})^{-1/3}. \tag{253}$$

Proposition 7.27 (Root-free no-drop cutoff)**.** *In the no-drop branch of Proposition 7.13, let*

$$Q = [A(q) + \tfrac{1}{4}]_< \in \{f-1, f\}, \qquad \mathcal{S}_n = D_A(q) + R_n.$$

Then

$$D_A(q) \geq \max\{0, C_0 K^{1/3} S_Q - Q - c_*\}. \tag{254}$$

Consequently the no-drop scalar is nonnegative whenever

$$K \geq K_{\mathrm{nd}}(n, Q) := \left(\frac{Q + n/2 + c_*}{C_0 S_Q}\right)^3. \tag{255}$$

Every branch below this cutoff is confined, for fixed (n, f, Q), to a compact strip with finitely many action walls.

Proof. For $0 \leq \theta \leq \pi/2$,

$$\sin\theta - \theta\cos\theta = \int_0^\theta t \sin t \, \mathrm{d}t \geq \frac{2\theta^3}{3\pi}.$$

Thus the angles in Corollary 7.4 satisfy

$$\frac{\pi}{2\theta_k} \geq C_0 K^{1/3} (k - \tfrac{1}{4})^{-1/3}.$$

The cap bound (30) is smaller than c_*, and the fractional-top term is nonnegative. Since the ball level count B is at least Q, this proves (254). Under (255) it gives $D_A(q) \geq n/2$, whereas (104) gives $R_n \geq -n/2$.

Finally, $K - \mu > \pi(f - 1/4)$ and $\mu \geq r + n$, so

$$K > \pi(f - \tfrac{1}{4}) + r + n.$$

Together with $K < K_{\mathrm{nd}}(n, Q)$ this bounds r, K, μ, every terminal level, and every sample index. On each open chamber the exact scalar has

$$\partial_K \mathcal{S}_n = \frac{2}{\pi} \int_r^K \sqrt{1 - z^2/K^2} \, \mathrm{d}z > 0, \qquad \partial_\mu \mathcal{S}_n = -\frac{2}{\pi} \int_r^\mu \sqrt{1 - z^2/\mu^2} \, \mathrm{d}z < 0.$$

It therefore reduces to the lower-K, upper-μ, and finitely many one-sided sample walls; the strict equality face is retained rather than passed through by continuity. ☐

Proposition 7.28 (Uniform half-unit terminal reserve)**.** *Let $q \leq \mu < q + 1$, $q \geq 3/2$, and $A(q) \geq 7/4$. Then, including at the lower ordinary-floor wall at q,*

$$\boxed{D_A(q) > \frac{1}{2}.} \tag{256}$$

Proof. Put

$$v = G_K(q), \qquad h = G_\mu(q), \qquad f = \left\lfloor A(q) + \frac{1}{4} \right\rfloor \geq 2, \quad Q = [A(q) + \tfrac{1}{4}]_<, \quad B = [v + \tfrac{1}{4}]_<.$$

If $q = \mu$, the inner cap vanishes. If $q < \mu$, strict convexity of G_μ on $[q, \mu]$ and $\mu - q < 1$ give

$$2 \int_q^\mu G_\mu(z) \, \mathrm{d}z < h(\mu - q) < h.$$

Moreover $h \le G_{q+1}(q)$, and $G_{q+1}(q)$ decreases with q. Outward arithmetic therefore gives the uniform cap bound

$$2\int_q^{\mu} G_\mu(z)\,\mathrm{d}z < G_{5/2}(3/2) < \frac{1}{5}. \tag{257}$$

For $1 \le k \le B$, let θ_k be the exact ball angle at level $k - 1/4$. Corollary 7.4, with its nonnegative fractional-top term omitted, gives

$$D_A(q) \ge \frac{\pi}{2}\sum_{k=1}^{B} \frac{1}{\theta_k} - Q - 2\int_q^{\mu} G_\mu(z)\,\mathrm{d}z. \tag{258}$$

If $q \ge K/2$, every retained level crosses strictly to the right of q, so $\theta_k < \arccos(q/K) \le \pi/3$. Away from the lower wall, $Q = f$ and $B \ge f$, which gives more than $f/2 - 1/5$. At the lower wall with $q < \mu$, one has $Q = f - 1$ but $B \ge f$; with $q = \mu$, the cap is zero and $B = Q = f - 1$, so pairing all $f - 1$ count units with their angle terms gives $D_A(q) > (f-1)/2 \ge 1/2$.

If $q < K/2$, let K_* be the unique solution of

$$G_{K_*}(3/2) = \frac{7}{4}.$$

Monotonicity gives $K \ge K_*$. At that radius, a directed 512-bit Arb calculation proves

$$\theta_1 < \frac{503}{500}, \qquad \theta_2 < \frac{11}{8},$$

and hence

$$\frac{\pi}{2}\left(\frac{1}{\theta_1} + \frac{1}{\theta_2}\right) - 2 > \frac{7}{10}, \qquad \frac{\pi}{2\theta_1} - 1 > \frac{1}{2}. \tag{259}$$

The angles decrease as K increases, and every later net level is positive. Away from the lower wall, the first inequality in (259) and (257) give more than $1/2$. At the lower wall the same two cases $q < \mu$ and $q = \mu$ are handled, respectively, by the extra ball level and the second inequality in (259). The computation is reproduced by `scripts/general_d_no_drop_round5_verify.py` and passes unchanged at 1024 bits. $\square$

Corollary 7.29 (Closure of the one-cell no-drop branch)**.** *Every no-drop branch with $f \ge 2$ and $n = 1$ satisfies*

$$\boxed{\mathcal{S}_1 = D_A(q) + R_1 > 0.} \tag{260}$$

Proof. Concavity of A on $[r, q]$ gives

$$2\int_r^q A(z)\,\mathrm{d}z \ge A(r) + A(q) \ge 2f - \frac{1}{2}.$$

Thus $R_1 \ge -1/2$, while Proposition 7.28 is strict. $\square$

Corollary 7.30 (A finite-chamber discard sector)**.** *For a no-drop branch with $f \ge 2$, put*

$$\varepsilon = A(q) + \frac{1}{4} - f, \qquad \Delta = A(r) - A(q).$$

If

$$2\varepsilon + \Delta + \frac{n\Delta}{3(2r+n)} \ge \frac{1}{2} - \frac{1}{2n}, \tag{261}$$

then $\mathcal{S}_n > 0$.

Proof. The scale-free shelf-curvature bound gives

$$R_n \geq \frac{n^2\Delta}{3(2r+n)} + n\left(2\varepsilon + \Delta - \frac{1}{2}\right) \geq -\frac{1}{2}.$$

Apply Proposition 7.28. $\square$

Theorem 7.31 (Quantitative no-drop transfer)**.** *Let $f \in \{2,3\}$, $b = f - 1/4$, and*

$$c_f = \frac{\sqrt{1+b^2}-1}{2}, \qquad C_K(f) = \left(\frac{f+1/2+c_*}{C_0 S_{f-1}}\right)^3, \qquad C_\delta(f) = \left(8\pi^2 f^2 C_K(f)\right)^{1/3}.$$

Suppose a residual no-drop candidate satisfies, with $\delta = K - \mu$,

$$\frac{c_f}{2} n \leq \delta \leq C_\delta(f) n, \qquad \frac{c_f^2}{8\pi^2(C_\delta(f)+2)} n^3 < K < C_K(f) n^3. \tag{262}$$

Put

$$s = \sqrt{\delta/K}, \qquad \kappa = s\delta = s^3 K, \qquad N = sn.$$

If $0 < s \leq 1/10$, then, including at every endpoint wall,

$$\boxed{s\mathcal{S}_n = s(D_A(q) + R_n) > \frac{1}{10}.} \tag{263}$$

Proof. Put $\alpha = \mu - q \in [0,1)$ and $X_q = s\alpha$. The residual endpoint floors give

$$b \leq A(q) < f \leq 3, \qquad \frac{21}{8} \leq \kappa. \tag{264}$$

The last inequality follows from the radius representation and $A(q) \leq 2\kappa/\pi$.

Use the exact head coordinate $X = s(\mu - z)$ and set

$$\mathcal{H}_{s,\kappa}(X) = A(\mu - X/s), \qquad H_\kappa(X) = \frac{c_0}{\sqrt{\kappa}}\left((\kappa+X)^{3/2} - X^{3/2}\right), \quad c_0 = \frac{2\sqrt{2}}{3\pi}.$$

Unlike a two-ball subtraction, the shell has the non-cancelling exact representation

$$\mathcal{H}_{s,\kappa}(X) = \frac{1}{\pi}\int_0^\kappa \frac{\sqrt{(\kappa+X-u)[2\kappa-(\kappa+X+u)s^2]}}{\kappa - us^2}\,\mathrm{d}u. \tag{265}$$

The ratio of its integrand to the integrand defining H_κ is

$$\mathcal{R}(u,X) = \frac{\sqrt{1-a(u,X)s^2}}{1-b_0(u)s^2}, \qquad a(u,X) = \frac{\kappa+X+u}{2\kappa}, \quad b_0(u) = \frac{u}{\kappa}.$$

For $0 \leq X \leq 2$ at every physical point,

$$0 \leq b_0 \leq 1, \qquad 0 \leq a \leq 1 + \frac{1}{\kappa} \leq \frac{29}{21}.$$

The inequalities

$$1 - y \leq \sqrt{1-y} \leq 1, \qquad \frac{1}{1-s^2} - 1 \leq \frac{100}{99}s^2$$

therefore give $|\mathcal{R}-1| \le (29/21)s^2$. At $X_q < s$, the sharper bound $a < 107/105$ gives

$$H_\kappa(X_q) \le \frac{\mathcal{H}_{s,\kappa}(X_q)}{1-(107/105)s^2} < \frac{31}{10}.$$

Since

$$0 < H_\kappa'(X) = \frac{3c_0}{2\sqrt{\kappa}}(\sqrt{\kappa+X}-\sqrt{X}) < \frac{1}{2},$$

one has $H_\kappa < 41/10$ on $[0,2]$. Integrating the ratio bound against the positive limiting integrand proves

$$|\mathcal{H}_{s,\kappa}(X) - H_\kappa(X)| < 6s^2 \qquad (0 \le X \le 2). \tag{266}$$

Put

$$\zeta = \frac{s}{2} + 6s^2 \le \frac{11}{100}, \qquad w = H_\kappa(0) = c_0\kappa.$$

The endpoint estimate, (266), and $H_\kappa' < 1/2$ give

$$w > b - \zeta. \tag{267}$$

For $X = \kappa u$, set

$$t = \sqrt{1+u} - \sqrt{u}, \qquad F(t) = \frac{3/t + t^3}{4}.$$

Then

$$\frac{H_\kappa(X)}{w} = F(t), \qquad H_\kappa'(X) = \frac{3c_0}{2}t.$$

Let X_* be the point at which $H_\kappa(X_*) = f + \zeta$. This point is positive: otherwise (266) at X_q would contradict $A(q) < f$. Furthermore

$$\frac{f+\zeta}{w} < \frac{f+\zeta}{f-1/4-\zeta} \le \frac{211}{164} < \frac{163}{125} = F(3/5).$$

Because F decreases, $t_* > 3/5$, and hence $H_\kappa' > 9c_0/10$ on $[0, X_*]$. With $d = f + \zeta - w$, this gives

$$0 < d < \frac{1}{4} + 2\zeta \le \frac{47}{100}, \qquad X_* < \frac{10d}{9c_0} < \frac{19}{10}.$$

Thus $X_q + X_* < 2$, so the error estimate applies on the whole possible negative head. Since $f+\zeta-H_\kappa$ is convex with endpoint values d and 0, its integral is at most $dX_*/2$. The exact increasing profile and the no-drop head identity therefore give

$$sR_n \ge -2\int_0^{X_*} (f+\zeta-H_\kappa(X))\,\mathrm{d}X \ge -\frac{10d^2}{9c_0}. \tag{268}$$

The first complete terminal level $\tau = 3/4$ is present because $G_K(q) \ge A(q) \ge 7/4$, including at the strict wall. Its angle lies in $(0, \pi/2)$. If

$$x_\tau = \left(\frac{3\pi\tau}{K}\right)^{1/3},$$

then $x_\tau^3 = 3(\sin\theta - \theta\cos\theta)$. The chord bound for sine and (264) imply $\theta < \sqrt{3}s$, while

$$\sin\theta - \theta\cos\theta \ge \frac{\theta^3}{3} - \frac{\theta^5}{30}$$

gives $x_\tau/\theta > 1 - 3s^2/10$. The exact-angle reserve, the cap bound, and $Q \le 3$ now give

$$sD_A(q) \ge \frac{\pi}{2\sqrt{2}} \left(\frac{b-\zeta}{3/4}\right)^{1/3} \left(1 - \frac{3s^2}{10}\right) - \frac{17}{5}s. \tag{269}$$

Here all further exact-angle and fractional-top terms were discarded in the favorable direction; the count and cap cost is less than $(3 + 2/5)s = 17s/5$.

Finally, all constants close rationally:

$$\frac{\pi}{2\sqrt{2}} > 1, \qquad c_0 > \frac{49}{165}, \qquad \frac{b-\zeta}{3/4} \ge \frac{164}{75} > \left(\frac{129}{100}\right)^3.$$

Adding (268) and (269) yields

$$s\mathcal{S}_n > \frac{129}{100}\frac{997}{1000} - \frac{17}{50} - \frac{10}{9}\frac{165}{49}\left(\frac{47}{100}\right)^2 = \frac{1758611}{14700000} > \frac{1}{10}.$$

At the lower ordinary-floor wall the separate identity (103) supplies one additional favorable unit for the original defect. The outward-rounded constants used above are replayed by `scripts/general_d_no_drop_round5_verify.py`; the certificate passes at both 512 and 1024 bits. □

Theorem 7.32 (Global finite exhaustion of no-drop data)**.** *Suppose $f \ge 2$, $n \ge 1$, and $\mathcal{S}_n < 0$. Then Proposition 7.27 must fail, and the following properties hold:*

1. *if $r \le K/2$, necessarily*
$$2 \le f \le 49, \qquad 2 \le n \le 48; \tag{270}$$
2. *if $r > K/2$ and $f \ge 4$, necessarily*
$$4 \le f \le 8, \qquad 2 \le n \le 39; \tag{271}$$
3. *in the two exceptional rows $f = 2, 3$, necessarily $2 \le n \le 379$. Before the quantitative transfer, an unbounded sequence could occur only in these rows, and then*
$$K - \mu \asymp n, \qquad K \asymp n^3, \qquad r/K \longrightarrow 1, \qquad \rho \longrightarrow 1. \tag{272}$$
Theorem 7.31 gives the explicit conclusion $n < 380$ in both rows. Thus the residual triples (n, f, Q) form a completely explicit finite set.

The certified central list has 647 necessary pairs after the two $n = 1$ pairs are removed; the outer $f \ge 4$ list has 60 necessary pairs and is unchanged. These counts precede certification of the continuous action chambers. This is a finite reduction, not a proof of the residual finite wall chambers.

Proof. Put

$$\delta = K - \mu, \quad \alpha = \mu - q \in [0,1), \quad b = f - \frac{1}{4}, \quad A(q) = b + \xi, \quad A(r) = b + \xi_0, \quad \Delta = \xi_0 - \xi.$$

Only $R_n < 0$ can remain. The exact shelf-curvature identity then gives

$$\xi_0 + \xi < \frac{1}{2}, \qquad 0 \le \xi < \frac{1}{4}, \qquad 0 \le \Delta < \frac{1}{2}, \qquad A(q) < f, \qquad A(r) < f + \frac{1}{4}. \tag{273}$$

In addition to $K < K_{\rm nd}(n,Q)$, three elementary action estimates are needed. The radius derivative and $K - q = \delta + \alpha < \delta + 1$ give

$$K \le U_f(\delta) := \frac{2\delta^2(\delta+1)}{\pi^2 b^2}. \tag{274}$$

The slope representation

$$\sigma(z) = \frac{1}{\pi}\int_\mu^K \frac{z}{a\sqrt{a^2-z^2}}\,\mathrm{d}a \ge \frac{\delta z}{\pi K^2}$$

and (273) imply

$$b\,n(2r+n) < K^2, \qquad b\,n(n+1) < K^2. \tag{275}$$

Finally, (69) with $p = n$ gives

$$\frac{\delta}{\pi} < f + \frac{1}{4} + \frac{r}{4n}. \tag{276}$$

If $r \le K/2$, then $r < \delta + n + 1$, and hence

$$\delta < D_{n,f} := \frac{f + 1/2 + 1/(4n)}{1/\pi - 1/(4n)}.$$

Therefore every residual central pair satisfies

$$b\,n(n+1) < U_f(D_{n,f})^2. \tag{277}$$

The left side increases with n and the right side decreases. Also

$$S_Q \ge \frac{3}{2}\left((Q+\tfrac{3}{4})^{2/3} - (\tfrac{3}{4})^{2/3}\right) \ge \frac{5}{4}Q^{2/3} \quad (Q \ge 7).$$

For $f \ge 16$ this first forces $n > f/6$; then $D_{n,f} < 5f$, $U_f(D_{n,f}) < 30f$, $n < 31\sqrt{f}$, and thus $f < 34596$. The remaining monotone endpoint comparisons are finite. Outward Arb arithmetic verifies, for both $Q = f-1, f$, that every $50 \le f \le 34595$ is excluded and leaves exactly 649 necessary pairs before the half-unit reserve is used. Its two $n = 1$ pairs are $(f,n) = (2,1), (3,1)$; Corollary 7.29 removes them and leaves 647 necessary pairs inside (270).

If $r > K/2$, the inequality $R_n < 0$ and monotonicity of σ give $\sigma(r) < 1/(2n)$. The radius representation also gives

$$\sigma(r) > \frac{\delta}{2\pi\sqrt{2K(\delta+n+1)}},$$

and therefore

$$K > \frac{\delta^2 n^2}{2\pi^2(\delta+n+1)}, \qquad \delta + 1 > c_f n, \quad c_f = \frac{\sqrt{1+b^2}-1}{2}. \tag{278}$$

With $\delta_{n,f} = \max\{\pi b, c_f n - 1\}$, every residual outer pair must satisfy

$$\frac{\delta_{n,f}^2 n^2}{2\pi^2(\delta_{n,f}+n+1)} < K_{\rm nd}(n,Q). \tag{279}$$

The same sum bound gives $f \le 66$. Once $c_f n - 1$ is active, the left side of the normalized comparison increases with n, while its right side decreases. Its limiting coefficients are

$$\gamma_f = \frac{c_f^2}{2\pi^2(c_f+1)}, \qquad \beta_J = \frac{3}{2\pi S_J^3}.$$

The certified signs $\gamma_2 < \beta_2$, $\gamma_3 < \beta_3$, and $\gamma_4 > \beta_3$ leave only $f = 2, 3$ as possible unbounded rows. For $4 \le f \le 66$, the finite monotone replay leaves exactly 60 necessary pairs in (271); none has $n = 1$.

For reproducibility, all outward-rounded comparisons just invoked are in `scripts/general_d_no_drop_exhaustion_verify.py`. At 256-bit precision it performs 138,184 central endpoint comparisons and 980 outer comparisons, and reports the pre-reserve counts 649 and 60 above. The same certificate was independently replayed through 1024 bits; the Round 5 replay confirms the updated counts 647 and 60.

It remains to rule out escape in $f = 2, 3$. On the upper half of $[\mu, K]$,

$$A(q) \ge \frac{\delta}{2\pi}\sqrt{\frac{\delta}{2K}}.$$

Together with $A(q) < f$, (255), and (278), this proves $\delta \asymp n$ and $K \asymp n^3$. Set

$$\eta = \frac{\delta}{K}, \qquad \kappa = K\eta^{3/2}, \qquad N = n\sqrt{\eta}.$$

This is exactly the critical scaling (114). For an explicit transfer, define $C_K(f)$ and $C_\delta(f)$ as in Theorem 7.31. If $n \ge \lceil 2/c_f \rceil$, then (278), the failed cutoff, and $\delta^3 < 8\pi^2 f^2 K$ give precisely

$$\frac{c_f}{2}n \le \delta \le C_\delta(f)n, \qquad \frac{c_f^2}{8\pi^2(C_\delta(f)+2)}n^3 < K < C_K(f)n^3.$$

The endpoint-action bound $A(q) \ge 2\kappa/(3\pi)$ and $A(q) < f$ give $\kappa < 3\pi f/2$. Since the cone also gives $\delta \ge c_f n/2$,

$$s = \frac{\kappa}{\delta} < \frac{3\pi f}{c_f n} < \frac{38}{n} \qquad (f = 2, 3). \tag{280}$$

Indeed, the two coefficients are $48\pi/(\sqrt{65} - 4)$ and $72\pi/(\sqrt{137} - 4)$, both strictly smaller than 38. Thus Theorem 7.31 excludes every residual candidate with $n \ge 380$, since then $s < 1/10$. Candidates below the preliminary threshold $\lceil 2/c_f \rceil$ already satisfy this bound. The two low-floor rows therefore satisfy the explicit bound $n < 380$. $\square$

8 A dimension-dependent no-mode region

The radial form (6), the one-dimensional Dirichlet Poincaré inequality, and $r^{-2} \ge 1$ on $(\rho, 1)$ give

$$\lambda_{\ell,n} \ge \frac{n^2\pi^2}{(1-\rho)^2} + \nu_{\ell,d}^2 - \frac{1}{4}.$$

Since $\nu_{0,d} = (d-2)/2$, one obtains the strengthened wall

$$N_D(A_{\rho,1}^{(d)}, K^2) = 0 \quad \text{if} \quad K^2 \le \frac{\pi^2}{(1-\rho)^2} + \frac{(d-2)^2 - 1}{4}. \tag{281}$$

The equality face belongs to the no-mode region because the counting function is strict.

In particular, for the general-dimensional Pólya theorem it is enough to prove the shifted-tail estimate in the universal active region

$$K > \frac{\pi}{1-\rho}. \tag{282}$$

Indeed, the complementary region has no eigenvalues already in dimension three, and the additional centrifugal term only enlarges it for $d > 3$. This observation does not prove the stronger standalone Conjecture 1.1 below the wall, but it removes that region from the spectral critical path.

9 Remaining proof obligations

The first interface-cell milestone is complete, and Proposition 6.8 proves that for each fixed ρ only a bounded frequency interval remains; in the thin sector Proposition 6.10 improves the sufficient threshold to $K \geq 125/[8(1-\rho)^2]$. Proposition 7.1 further replaces the many-cell dynamics by the single scalar (66). For the spectral theorem this scalar is needed only in (282). Lemma 7.2 sharpens its head term, while Corollary 7.4 replaces qualitative terminal nonnegativity by a fractional-top positive sum, and Corollary 7.5 covers the empty-level case. Proposition 7.15 closes an explicit high-floor finite-shell sector, while Theorem 7.16 rules out a negative leading obstruction on the grazing ray. For $p = n$, Proposition 7.13 identifies the exact endpoint phase and Theorem 7.32 proves a global finite reduction. It does not certify every member of that finite family. Theorems 7.10 and 7.18 remove every noncompact first-drop escape. Theorem 7.12, together with Theorems 7.6– 7.8, closes every ordinary first-floor active wall; (281) removes the below-active region from the spectral critical path. Theorem 7.19 gives a first explicit high-floor reduction, and Theorem 7.20 supersedes its enormous cutoff: it first reduced every negative candidate to the Round 7 box (142)– (143). The extended Theorem 7.21 next supersedes that box, and the terminally strengthened Theorem 7.23 supersedes it again, Theorem 7.25 sharpens it once more, and Theorem 7.26 gives the current sharpest box: every negative candidate lies in (235)– (236). This sharper reduction still does not certify the residual compact walls. The primary route now has the following precise steps.

1. In the no-drop branch, certify the residual one-sided walls in the explicit central and outer boxes (270)–(271), together with the two rows $f = 2, 3$, $2 \leq n \leq 379$ from Theorem 7.31. The $n = 1$ branch is already closed by Corollary 7.29; in the remaining chambers the discard sector (261) and the favorable endpoint unit in (103) should be retained.

2. In the high-floor $p < n$ branch, Theorem 7.20 historically closes $\rho > 99/100$, and Theorem 7.21 extends the exclusion to $\rho \geq 603/625$, and Theorem 7.23 extends it to $\rho \geq 477/500$, and Theorem 7.25 uses the active width and sharp terminal cap to extend it further to $\rho \geq 93/100$, and Theorem 7.26 combines the automatic $q \geq 5/2$ cap with a sharper absent-level chord to reach $\rho \geq 927/1000$. It leaves only

$$\frac{1}{4626882} < \rho < \frac{927}{1000}, \qquad \frac{21}{4} < K < 2313441,$$

   with the exact integer ranges in (236). The fixed-ratio compact analytic walls inside those bounds are still uncertified; they require sharper analytic localization or directed-rounding certificates, not a claim of closure from finiteness alone. There is no remaining ordinary first-floor wall.

3. Insert the resulting all-frequency shifted-tail theorem into Theorem 5.3. No dimension-dependent angular or optical theorem is then needed.

If the first step fails only for an explicitly controlled family of tails, the exact identity (24) is the appropriate aggregate fallback: the weighted negative part must be compared directly with $\mathcal{B}_d(A)$.

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